

Data-Enabled Mental Health: Exploring ML Models for Screening and Predicting the Severity of Depression

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Data-Enabled Mental Health: Exploring ML Models for Screening and Predicting the Severity of Depression

Abstract

Depression is a pervasive and debilitating mental health disorder with a significant global impact, affecting approximately 280 million individuals and contributing to over 720,000 suicides annually. Epidemiological evidence indicates that women are affected about 50% more frequently than men (**World Health Organization, 2017; WHO, 2023**). Early detection is essential for effective intervention; however, prevailing screening methods, such as self-administered instruments (e.g., PHQ-9) and clinician-led assessments, are resource-intensive and susceptible to subjective bias. Advances in artificial intelligence (AI) present a promising avenue for improving screening by passively analysing digital behavioural data, including linguistic patterns from social media, speech characteristics, and smartphone sensor metrics, to identify indicators of psychological distress (**Roberts et al., 2025**). Nevertheless, many existing AI models function as opaque “black box” systems, delivering only binary classifications without transparent reasoning or severity estimation (**Liu et al., 2022**), which limits their trustworthiness and applicability in clinical practice.

This study addresses these limitations by implementing and evaluating an interpretable machine learning approach for depression screening that moves beyond binary classification to predict clinically relevant multiclass severity levels. Using longitudinal data from the National Health and Nutrition Examination Survey (NHANES) with repeated Patient Health Questionnaire-9 (PHQ-9) assessments, two clinically meaningful target variables (Depression_Severity and Final_Risk) were engineered. A suite of models was implemented, including interpretable Generalised Additive Models (GAMs) and Explainable Boosting Machines (EBMs), alongside high-performance black-box models such as Random Forest, XGBoost, and Long Short-Term Memory (LSTM) networks. Models were trained and evaluated using Group Shuffle Split and Stratified Group K-Fold cross-validation. Performance was measured using Accuracy, F1-score, and Area Under the Receiver Operating Characteristic Curve (ROC-AUC), with statistical significance assessed via DeLong’s and McNemar’s tests. Interpretability was analysed using SHAP (SHapley Additive exPlanations) to identify and visualise key predictive features. Results show that while black-box models achieved slightly higher accuracy, the adapted multiclass GAMs delivered a strong balance of interpretability and performance, providing a clinically transparent and operationally viable solution. The project culminated in PsyCheck, a semi-functional web-based prototype for user-centred depression screening and monitoring, evaluated through structured usability testing.

Keywords: Machine Learning, Interpretable AI, Depression Screening, Mental Health, Generalised Additive Models (GAM), Explainable Boosting Machine (EBM), SHAP, Longitudinal Data, Digital Phenotyping, Clinical Decision Support.

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Abbreviations

AI	Artificial Intelligence
DSM-5	Diagnostic And Statistical Manual of Mental Disorders, 5th Ed.
GAM	Generalized Additive Model
GDPR	General Data Protection Regulation
ML	Machine Learning
NHANES	National Health and Nutrition Examination Survey
NHIS	National Health Interview Survey
PHQ-9	Patient Health Questionnaire-9
PLES	Professional, Legal, Ethical, And Social
ROC-AUC	Receiver Operating Characteristic - Area Under the Curve
WHO	World Health Organization

1 Introduction and Rationale behind Proposal

Depression is a widespread and severe mental disorder with significant global health implications, affecting more than 300 million people and ranking among the leading causes of disability worldwide **(WHO, 2023; American Psychiatric Association, 2013)**. Clinically, major depressive disorder is characterised by persistent low mood and loss of interest or pleasure, accompanied by symptoms such as impaired concentration, excessive guilt or low self-worth, disturbed sleep, appetite or weight changes, fatigue, and recurrent thoughts of death or suicide **(WHO, 2017)**. For diagnosis, these symptoms must persist for at least two weeks and cause substantial functional impairment. Standard diagnostic criteria, such as those outlined in the DSM-5, require structured clinician-led interviews and assessments **(American Psychiatric Association, 2013)** in which patients are evaluated for both the presence and severity of depression.

Several validated instruments support screening and diagnosis. The Patient Health Questionnaire-9 (PHQ-9) is among the most widely used, comprising nine self-report items aligned with the DSM-5 symptom criteria. Patients rate symptom frequency over the past two weeks to produce a score from 0 to 27, with cut-off points defining mild, moderate, moderately severe, and severe depression. A score of ≥ 10 yields approximately 88% sensitivity and 88% specificity for detecting major depression in primary care **(Kroenke et al., 2001)**. The PHQ-9's brevity, reliability, and ease of administration have led to its widespread adoption as a first-line screening tool **(Kroenke et al., 2001; World Health Organization, 2017)**. Other rating scales, including the Beck Depression Inventory and the Hamilton Rating Scale for Depression, are also used, but the PHQ-9's practicality makes it the preferred choice in many primary care settings **(Beck et al., 1996)**.

Despite their utility, existing approaches face significant challenges. They rely heavily on subjective self-report and clinician judgment, making them susceptible to under- or over-reporting due to stigma, recall bias, or current mood. Clinician-administered assessments are resource-intensive and subject to variability in interview style and context. Under-diagnosis remains common, with only about half the patients with major depression correctly identified in primary care **(Mitchell et al., 2009)**. Many individuals do not seek help or present with non-specific physical complaints that mask mood-related symptoms, resulting in large numbers remaining undiagnosed and untreated, thus resulting in greater functional impairment and higher suicide risk. Using it for routine screening on a larger scale is further constrained by the limited capacity of healthcare systems, particularly in resource-limited settings.

These limitations have inspired data-enabled approaches that can detect and monitor depression more proactively. Benefitting from the growing availability of digital data, such as social media activity, smartphone sensor logs, and voice recordings, these methods aim to passively identify behavioural markers of depression on a larger scale.

This approach, known as digital phenotyping, uses everyday digital behaviours as proxies for mental health status (**Roberts et al., 2025**). Such methods could complement traditional screening by identifying at-risk individuals who might otherwise remain undetected, enabling earlier diagnosis and intervention (**Torous and Roberts, 2017**). Extending detection beyond clinical settings into real-world contexts offers the potential to reduce the burden on healthcare systems and address critical gaps in care. This study sought to advance this vision by developing an interpretable machine learning approach for predicting depression severity levels, prioritising both predictive accuracy and ethical deployment.

1.1 Research Question

Primary Research Question:

How can machine learning models be applied to longitudinal PHQ-9 data to accurately and interpretably predict depression severity and final risk and compare two distinct training-validation approaches, to address the limitations of current depression screening methods?

Sub-Questions:

1. Severity Prediction:

How can longitudinal PHQ-9 responses be transformed into clinically meaningful multiclass severity categories to enable accurate model training and evaluation?

2. Final Risk Assessment:

How can a composite “final risk” outcome be defined and predicted to combine severity classification with an assessment of progression or persistence risk?

3. Two-Method Validation Approach:

How do Group Shuffle Split and Stratified Group K-Fold cross-validation strategies affect model performance, robustness, and generalisability when predicting severity and final risk?

4. Model Performance and Comparison:

How do interpretable models, such as Generalised Additive Models (GAMs) and Explainable Boosting Machines (EBMs), compare with black-box models in predicting severity and final risk, in terms of accuracy, F1-score, and ROC-AUC?

5. Interpretability and Clinical Insight:

How can SHAP techniques be used to visualise the features influencing severity and risk predictions, in a clinically meaningful way?

1.2 Aim

The aim of this study is to develop and evaluate a data-driven depression screening approach that uses machine learning to predict an individual's depression severity and final risk in a transparent, clinically interpretable manner. This project seeks to address key shortcomings in existing screening methods by creating a model that can reliably discriminate between severity levels, explain its reasoning, and bridge the gap between coarse screening and detailed clinical assessment.

1.3 Objectives

To achieve the above aim, the project will pursue the following objectives:

- 1.** To validate exploratory findings of the NHANES longitudinal PHQ-9 dataset through statistical analysis, achieving $p < 0.05$ significance for at least 80% of tested relationships of the engineered clinically relevant multiclass target labels (Depression Severity and Final Risk).
- 2.** To train two categories of models, interpretable GAM variants and selected black-box models, using Group Shuffle Split and Group K-Fold cross-validation, producing training performance metrics with accuracy above 60%, along with corresponding F1-scores and ROC-AUC values for all model training instances.
- 3.** To compare the model performance of interpretable GAM variants and black-box models by determining statistical significance of differences at $p < 0.05$ using DeLong's and McNemar's tests.
- 4.** To analyse the interpretability of the top-performing models using SHAP, to obtain the top five predictive features and their contribution to severity and risk predictions through summary plots and feature effect visualisations.
- 5.** To develop a semi-functional front-end prototype of the PsyCheck Web application to conduct structured UI/UX usability testing with at least 10 participants, achieving a System Usability Scale (SUS) score above 80 and qualitative feedback on interface clarity and trustworthiness.

By achieving these objectives, this project will contribute an interpretable machine learning solution for data-enabled depression screening that combines predictive accuracy with clinical applicability. The work aims to define a viable pathway for scalable and timely identification of depression cases, incorporating severity estimation, using AI in a manner that enhances clinical decision-making while addressing the recognised limitations of both manual diagnostic processes and existing algorithmic methods.

1.4 Contributions

This project makes the following key contributions to the field of data-driven mental health screening:

- **Multiclass Severity Prediction:** Whereas much of the existing research focuses on binary classification (depressed vs. not depressed), this study successfully implemented and validated models for multiclass severity and risk prediction. This provides a more nuanced output that is better aligned with clinical practice, where the level of depression dictates the treatment pathway.
- **Adaptation of GAMs for Multiclass Classification:** The standard pygam library is designed for binary classification or regression. A novel contribution of this project was the implementation of custom One-vs-Rest (OVR) wrapper classes (LinearGAMOVR, LogisticGAMOVR), which successfully adapted the highly interpretable GAM framework for this multiclass, longitudinal classification task.
- **Rigorous Comparative Framework:** The project provides a comprehensive, head-to-head comparison of interpretable "glass-box" models against high-performance "black-box" models within a robust cross-validation framework (Stratified Group K-Fold). The statistical validation using DeLong's and McNemar's tests offers a rigorous methodology for assessing not only whether the models perform differently, but also how.
- **Emphasis on Longitudinal Feature Importance:** Through SHAP analysis, the study statistically validated the clinical intuition that recent symptom patterns, particularly functional impairment (DPQ10), are the most critical predictors of near-future depression severity. This highlights the value of longitudinal data over single, cross-sectional assessments.
- **From Model to Prototype:** The project successfully bridged the gap between theoretical modelling and practical application by developing the 'PsyCheck' prototype, demonstrating how such an interpretable screening tool could be deployed in a user-facing application.

2 Literature Review

2.1 Background on Depression Screening and Diagnostics

Globally, depression affects an estimated 280 million people and remains a leading cause of disability (**World Health Organization, 2023**). It can profoundly impact mood, behaviour, and daily functioning, and in severe cases lead to suicide. Despite the availability of effective treatments, a large proportion of those suffering – especially in low-resource settings – do not receive care. Over 75% of people with depression in many low- and middle-income countries get no treatment for their condition due to factors like lack of providers and stigma (**World Health Organization, 2023**). Early identification and intervention are therefore critical to reduce the burden of untreated depression.

In clinical practice, depression is diagnosed based on interviews and standardized criteria such as those in the DSM-5, administered by mental health professionals. To facilitate depression screening in primary care and research, several questionnaire-based instruments have been developed. As discussed, the PHQ-9 is one of the most widely used tools, covering the DSM-5 symptom criteria for major depression in nine self-report items (**Kroenke et al., 2001**). Patients rate each symptom's frequency over the past two weeks, yielding a score from 0 to 27. The PHQ-9 is a reliable indicator of depression severity; a score of 10 or above is often used as a threshold for clinically significant depression, with a sensitivity and specificity of around 88% for major depressive disorder (**Kroenke et al., 2001**). Its brevity and proven validity have led to broad adoption of the PHQ-9 in both clinical settings and large-scale screenings. Other scales, such as the Beck Depression Inventory-II and the Hamilton Rating Scale for Depression, are also employed to assess symptom severity. The Beck Depression Inventory (BDI-II), for example, is a longer self-report measure of depressive symptoms (**Beck et al., 1996**). However, due to its simplicity and ease of use, the PHQ-9 is especially popular for initial screening, particularly in primary care environments.

Traditional screening tools like the PHQ-9 rely on individuals to self-report symptoms or on clinicians to administer questionnaires. This process can be time-consuming and is subject to various biases (e.g., recall bias on the patient's side or interviewer bias on the clinician's side). Moreover, many people with depression do not actively seek help or may downplay their symptoms during clinical assessments. (**Torous and Roberts, 2017**).

These challenges in relying solely on patient self-report and clinician observation have opened the door for data-driven methods that can potentially detect depression more objectively and proactively. By using data collected from individuals' day-to-day lives, these methods aim to identify depression-related patterns without waiting for the person to report their symptoms. In summary, while established clinical instruments lay the foundation for depression diagnostics, there is a clear need for innovative screening methods that can reach populations more broadly and thus, reduce the burden on healthcare systems (Torous and Roberts, 2017).

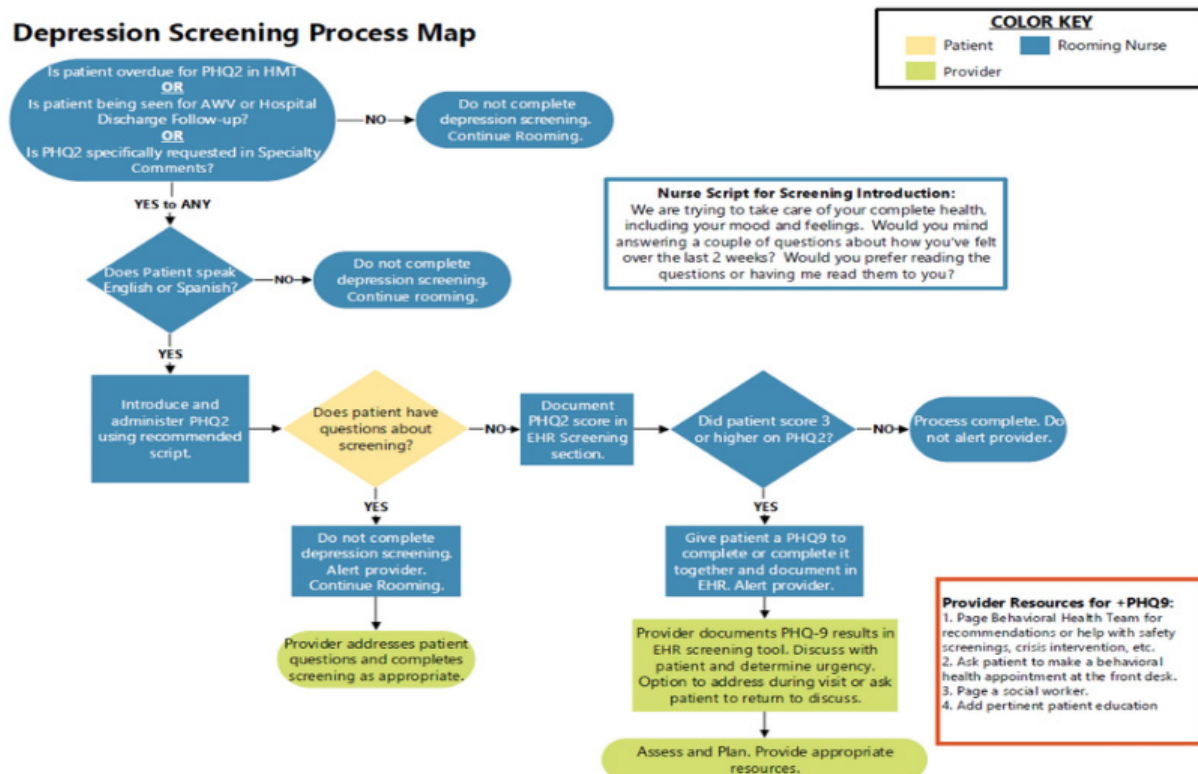


Fig. 2.1- Depression Screening Clinical Process (Source: Blackstone et al., 2022)

Depression Screening Tools					
TOOLS	Beck Depression Inventory	PHQ-9	DASS-21	Hamilton Depression Rating	CES-D
STRENGTH	Comprehensive, with a broader range of symptom assessment.	Easy and quick to administer, well-validated in diverse populations.	Assesses multiple emotional states, quick to administer	Detailed and thorough, excellent for clinical trials.	Useful for large scale studies, easy to administer
WEAKNESS	Longer to administer, not feasible in fast-paced settings.	May not capture all nuances of depressive symptoms.	Less specific to depression alone	Requires significant training, less practical for primary care.	Less specificity for clinical diagnosis

Fig. 2.2- A Comparison of Depression Screening Tools (Source: Thanvi, 2024)

2.2 Data-Driven Approaches to Depression Detection

In recent years, researchers have begun applying data-driven techniques to mental health assessment, inaugurating the field of “digital psychiatry.” The premise is that a person’s digital footprints and other objective data sources can reveal subtle indicators of their mental well-being. A variety of data modalities have been explored for depression detection:

- **Social media and Textual Data:** Individuals often express their feelings and thoughts on social media platforms, making these texts a rich source of signals for mental health. Studies have used natural language processing to analyse social media posts for linguistic markers of depression. For instance, **(Aldarwish and Ahmad , 2017)** demonstrated that features from Twitter posts can be used to predict users’ depression levels. Similarly, **(Shen et al., 2017)** proposed a multimodal approach combining textual and visual content from social media to detect depression, achieving promising accuracy in identifying depressed individuals. These studies show that language use (e.g., frequency of negative emotion words, first-person pronouns) and even images shared online provide hints to a person’s mental state.
- **Smartphone Sensor Data:** Modern smartphones continuously collect data on location, movement, activity levels, and social communication. Researchers have correlated passive sensor data with depression severity. For example, **(Saeb et al., 2015)** found that GPS movement patterns and phone usage statistics were significantly associated with depressive symptom severity in daily life. Reduced mobility, irregular routines, and diminished social communication detected via phone sensors were indicators of higher depression scores. These findings suggest that smartphones can serve as passive detectors of behavioural changes relevant to mental health.
- **Speech and Audio:** Changes in speech patterns and voice can also reflect depression – such as slower speech, longer pauses, or reduced variation in tone. Some studies (e.g., the Distress Analysis Interview Corpus) have examined recorded interviews to detect depression from vocal cues. The Distress Analysis Interview Corpus (DAIC) contains recordings of clinical interviews conducted by virtual agents and has been used to train models to recognize depression from speech and facial expressions **(Gratch et al., 2014)**. Such multimodal datasets enable analysis of audio (what is said and how it is said) and visual cues (e.g., facial expression, eye contact) for signs of mental distress.

- **Multimodal Fusion:** Combining multiple data sources can improve detection accuracy, as different modalities capture different aspects of depression. Recent research has explored multimodal machine learning models that integrate text, audio, video, and sensor data. For example, **(Tzirakis et al., 2018)** developed an end-to-end deep neural network that fuses audio and visual inputs for emotion recognition, which is relevant to detect depression-related affective states. In the context of depression, **(Haque et al., 2018)** built a deep multimodal neural network to detect mental disorders using a combination of textual and acoustic features, showing that multi-layer perceptron models can integrate signals from different channels to improve prediction. A scoping review by **(Mlakar et al., 2022)** indicates that multimodal classification of depression (using combinations of modalities like text, audio, video, and physiological signals) often outperforms single-modality approaches, though many existing models are still in experimental stages.
- **Early Risk Prediction:** Beyond cross-sectional screening, there is an interest in detecting depression as early as possible, before full-blown clinical manifestation or crisis. The CLEF eRisk challenges have provided benchmarks for early prediction of depression from forum posts and other sequential data. **(Losada, Crestani and Parapar , 2017)** introduced the first eRisk challenge in 2017, which spurred the development of algorithms that process a user's posts over time and attempt to detect early signs of depression. Subsequent research built on this, applying advanced natural language processing methods. For instance, transformer-based language models have been employed to improve early detection: **(Martínez-Castaño et al., 2021)** explored BERT-based transformers for early detection of mental health issues in the eRisk context. Likewise, **(Matero et. al., 2019)** demonstrated that using BERT with additional context (user history, hierarchy of posts) could enhance the identification of users at risk of suicide on social media. These studies illustrate the shift toward utilizing state-of-the-art language models in mental health, given their success in other domains. (Notably, the transformer architecture underpinning models like BERT was introduced by **(Vaswani et al., 2017)**, and its self-attention mechanism has proven effective in modelling language relevant to mental health signals.)

The overall performance of AI models in depression detection has been encouraging but still leaves room for improvement. (Liu *et al.*, 2022) Showed that some models report moderate accuracy – for example, certain social media-based classifiers achieve F1-scores around 0.7 for detecting depression. However, these approaches often focus on binary classification (depressed vs. not depressed) and do not provide estimates of depression severity. Moreover, many are “black boxes” that offer little insight into the reason behind a prediction, which can be problematic for clinical acceptance. These observations highlight a gap that the current project aims to address by developing models that can output nuanced severity levels and are interpretable to stakeholders.

2.3 Digital Mental Health Tools and User Engagement

It is important to contextualize AI-based screening within the broader landscape of digital mental health interventions. Numerous smartphone applications and digital tools have been developed to help manage or treat depression and related conditions. For example, fully automated conversational agents (chatbots) like **Woebot** and **Wysa** engage users in therapeutic conversations. A randomized controlled trial showed that **Woebot**, a chatbot delivering cognitive-behavioural therapy, could significantly reduce symptoms of depression and anxiety in young adults (Fitzpatrick *et al.*, 2017). Similarly, real-world evaluations of **Wysa** (an AI-based chatbot for mental well-being) indicated user-reported improvements in depressive symptoms (Inkster *et al.*, 2018).

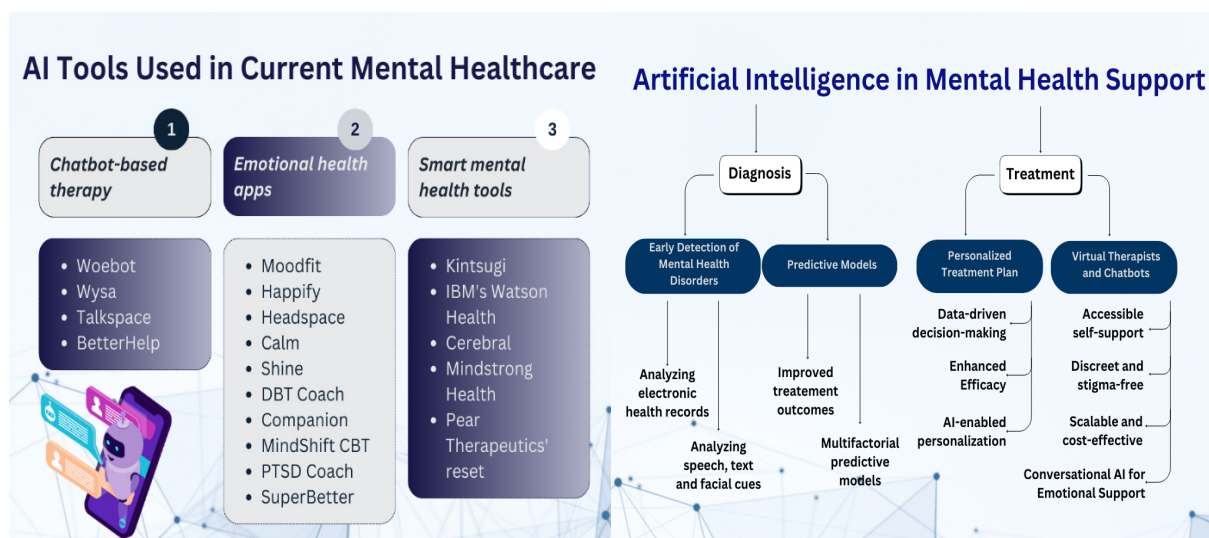


Fig. 2.3- Existing Tools in a Clinical Setting (Source: Enago, 2024)

Beyond chatbots, many mental health apps focus on mindfulness and mood monitoring. (Bostock *et. al.*, 2019) found that regular use of a mindfulness meditation app led to reductions in work stress and improvements in well-being.

Another trial comparing popular mindfulness apps observed significant mental health benefits for users (**Flett et. al., 2018**).

A recent meta-analysis of mindfulness-based interventions delivered via technology confirmed their efficacy across various psychiatric disorders (**Goldberg et. al., 2022**).

While these digital interventions show promise, user engagement remains a major challenge. Many people download wellness apps but fail to stick with them. A systematic review by (**Borghouts et. al., 2021**) identified common barriers to user engagement with digital mental health tools, including lack of personalization, technical difficulties, and concerns about privacy. The content quality and evidence base of mental health apps also vary widely.

(**Terhorst et al. , 2021**) conducted a comprehensive evaluation of depression-related apps and found that while some scored high on usability and information quality, many others had significant deficiencies (e.g., few evidence-based features or poor security). These findings underscore that simply having a digital tool is not enough – its design, credibility, and user experience are crucial for real-world impact. For an AI-based screening tool to be effective, it must not only be accurate but also engaging and trustworthy to its users.

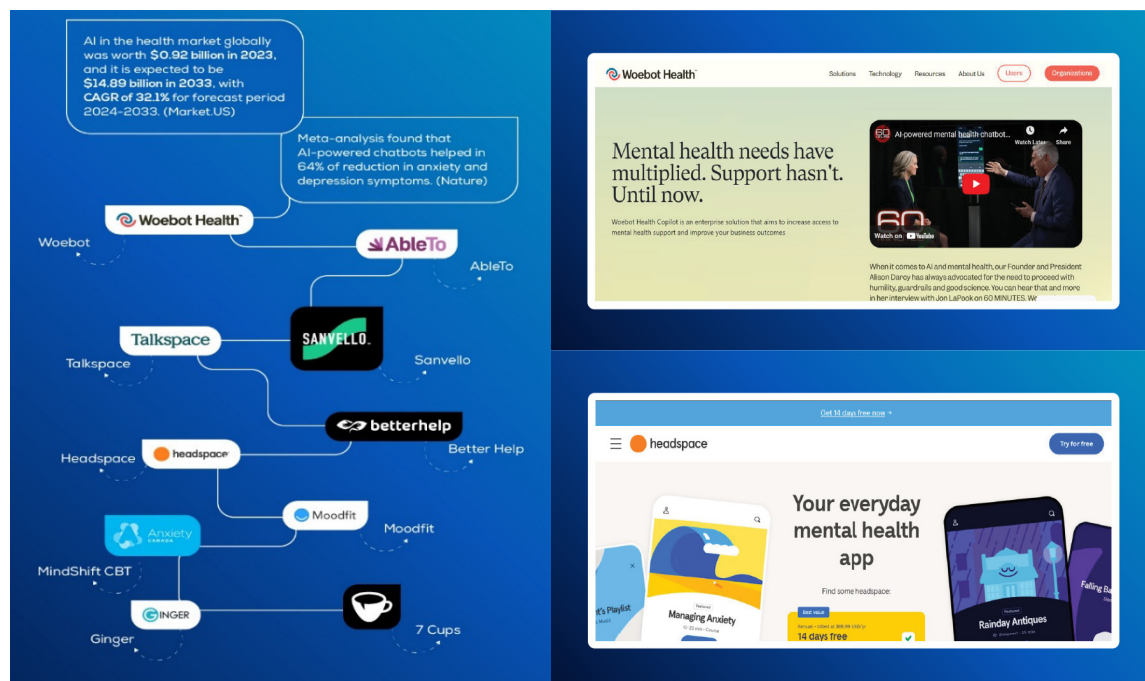


Fig. 2.4 - Existing Tools in a Clinical Setting (Source: Verma, 2024)

2.4 Ethical, Privacy, and Bias Concerns in AI Screening

Deploying AI in mental health screening raises important ethical and practical considerations. Data privacy is paramount when dealing with sensitive personal information like mental health status. Any system that analyses user data (be it social media content or sensor logs) must ensure compliance with privacy laws and secure handling of data. In addition, algorithmic fairness is a concern: models should be evaluated for potential biases in performance across different demographic groups to avoid worsening healthcare disparities. For instance, an AI model might perform differently for men vs. women or younger vs. older individuals if those differences are present in the training data. Identifying and correcting such biases is critical before clinical use. Another consideration is the **transparency and interpretability** of AI decisions. Unlike a human clinician's rationale, AI predictions can be opaque. Black-box algorithms that cannot explain their reasoning which may undermine user trust and make it difficult for clinicians to validate or act on the results.

There is an increasing call for “glass box” models in high-stakes fields like healthcare, where interpretability is valued alongside accuracy (**Torous and Roberts, 2017**). Interpretable models can provide explanations (for example, indicating which inputs most influenced a decision), which can help both clinicians and patients understand and trust the system's output.

The public's perception of AI in mental health is also an important factor. People may be wary of machines judging their mental health or might not trust recommendations that come from an algorithm. A recent study surveying public perspectives on AI-based mental health diagnostics highlighted both interest and concern – participants appreciated the potential for early help but raised ethical questions about accuracy and empathy (**Stanley et al., 2024**). Clear communication about what an AI tool can and cannot do, and ensuring it is used to assist rather than replace professional judgment, are key to ethical deployment (**Warrier et al., 2023**).

2.5 Updated Literature Review

2.5.1 Model Selection: Balancing Accuracy and Interpretability in Clinical AI

The trade-off between predictive accuracy and interpretability remains central in clinical machine learning research. Complex, non-linear “black-box” models, such as deep neural networks and gradient-boosted trees, often achieve high predictive accuracy but are criticised for their opacity. This lack of transparency can undermine trust and hinder adoption in clinical environments where practitioners are ethically and professionally obliged to understand the rationale behind diagnostic or prognostic indicators (**Adadi and Berrada, 2018**).

In contrast, “glass-box” models such as **Generalised Additive Models** (GAMs) and **Explainable Boosting Machines** (EBMs) offer inherent intelligibility. (**Caruana et al., 2015**) demonstrated that GAMs could match the accuracy of a leading black-box model for pneumonia risk prediction while revealing clinically meaningful but counter-intuitive patterns in the data. This project contributes to this debate by directly comparing GAMs and EBMs with high-performance black-box models using a longitudinal PHQ-9 dataset. Unlike studies that optimise solely for a single performance metric, this study examines the practical trade-offs between accuracy and interpretability in the context of depression severity and risk prediction.

2.5.2 Validation Strategy: Ensuring Robustness with Group-Aware Methods

The validity of clinical prediction models depends on robust evaluation strategies. In longitudinal datasets, multiple observations are nested within individuals, and standard record-wise k-fold cross-validation can lead to data leakage where data from the same individual appears in both training and test sets. , resulting in inflated performance estimates (**Saeb et al., 2017**).

To address this, the literature advocates for group-aware validation methods. This project employs two such approaches: **Group Shuffle Split**, which ensures that all records for a given patient remain in a single fold, and **Stratified Group K-Fold**, which additionally preserves class proportions in each split. This dual-method strategy aligns with best practice recommendations and more accurately simulates real-world clinical deployment, improving the reliability of generalisation estimates.

2.5.3 Hyperparameter Optimisation in Clinical AI

Hyperparameter optimisation (HPO) can significantly influence model performance, yet many clinical AI studies fail to report their tuning strategy, instead relying on default settings

(Cabitza et al., 2023). This omission reduces reproducibility, and casts doubt on reported performance. In contrast, this project integrates an explicit HPO process into its modelling pipeline.

For interpretable models such as GAMs, a grid search was performed over spline counts and regularisation parameters (`n_splines_grid`, `lam_grid`) within each fold of the grouped cross-validation scheme to prevent leakage.

Black-box models were tuned over learning rates, depth parameters, and ensemble sizes within similarly controlled folds. This ensures that performance comparisons between models are fair, optimised, and replicable.

2.5.4 Statistical Significance Testing for Comparative Evaluation

Performance metrics alone cannot determine whether observed differences between models are meaningful. Differences in accuracy or ROC-AUC can arise by chance, particularly in smaller datasets.

Statistical methods such as McNemar's test **(McNemar, 1947)** for classification errors and DeLong's test for comparing correlated ROC curves **(DeLong, DeLong and Clarke-Pearson, 1988)** are well-established in clinical model evaluation.

Many applied AI studies omit these tests, but the evaluation pipeline in this study integrates both. It enabled the researchers to make statistically grounded claims, for example by confirming the statistical significance of the difference between a marginally higher ROC-AUC for XGBoost and GAM, thereby supporting informed model selection.

2.5.5 Interpretability, Clinician Trust, and End-User Understanding

Adoption of clinical AI tools ultimately hinges on trust. Clinicians often prefer inherently interpretable models over post-hoc explanations of black-box systems (Holzinger et al., 2019). Furthermore, explanation usability literature indicates that interpretability must be actionable and comprehensible for its intended users. This project addresses both issues. First, it prioritises interpretable models (GAMs, EBM) as core architectures.

Second, it employs SHAP (Lundberg and Lee, 2017) to generate feature importance insights that are presented in clinically meaningful terms, such as the finding that functional impairment (PHQ-9 Item 10) in the most recent week is the strongest predictor of near-term risk. Coupled with the PsyCheck prototype, this ensures the bridge from algorithm output to clinician decision-making is clear and credible.

2.5.6 Usability Studies: Bridging Algorithm Performance and Real-World Adoption

A clinically accurate algorithm has little value if patients or clinicians find it unusable or untrustworthy. Human–Computer Interaction (HCI) research consistently emphasises that digital health interventions must incorporate user-centred design and formal usability evaluation to succeed in practice (Lattie et al., 2019). Poor usability, unclear feedback, and non-intuitive interfaces are the primary reasons frequently cited for low engagement and high attrition in mental health apps (Borghouts et al., 2021).

The development of the PsyCheck prototype and its associated usability survey in this project reflects these principles. Our process aligns with the “evidence-based design” paradigm (Inkster et al., 2018), involving iterative design informed by end-user feedback. By collecting both quantitative usability metrics (System Usability Scale) and qualitative feedback, the researchers tested the tool for clinical and practical viability, a step often missing in purely theoretical ML studies.

3 Requirements Analysis and Methodology

This chapter outlines the methodological framework of the project, beginning with the functional and non-functional requirements that guided the development of the PsyCheck screening tool. It then describes the dataset, feature engineering, data preprocessing steps, model development approach, software and hardware resources, and the comprehensive evaluation plan used to assess and compare the machine learning models. The chapter concludes with the project timeline and a risk analysis.

3.1 Requirements Analysis

From the project's aim and objectives, we derive specific requirements for the depression severity screening system. To ensure that the final system was technically proficient, clinically relevant, and usable, a set of functional and non-functional requirements were defined. We assume a general self-screening scenario (e.g., via a smartphone/computer web interface) with potential for later clinical integration.

3.1.1 Functional Requirements (FRs) :-

- **FR1: Daily Journaling with AI-Powered Sentiment Analysis :-**

The system must provide an interface for users to write and save daily journal entries. Each entry is analysed by an AI model (Google's Genkit) to generate a sentiment score and brief qualitative analysis, enabling users to reflect on their emotional state.

- **FR2: Interactive PHQ-9 Assessment via Chatbot :-**

The system must administer the nine-item Patient Health Questionnaire (PHQ-9) using a conversational chatbot interface. The chatbot guides the user through each question, records responses, calculates the final score, and provides an automated severity-based intervention message.

Proposed Treatment Action by PHQ 9 Score		
PHQ-9 Score	Depression Severity	Proposed Treatment Actions
0-4	Non – Minimal	None
5-9	Mild	Watchful waiting; repeat PHQ 9 at follow-up
10-14	Moderate	Review treatment plan if not improving in past 4 weeks; Consider discussion of additional support such as pharmacotherapy
15-19	Moderately Severe	Consider adjusting treatment plan and/or frequency of sessions; Discuss additional supports such as pharmacotherapy; For SonderMind Anytime Messaging clients, consider converting from asynchronous to synchronous therapy channels
20-27	Severe	Adjust treatment plan; focused assessment of safety plan and pharmacotherapy evaluation/ re-evaluation; If emergent then refer to higher level of care; Likely Not a candidate for asynchronous/text therapy

Table 3.1- PHQ-9 Depression Screening Tool in Clinical Settings (Source: SonderMind, 2023)

- **FR3: Lifestyle Goal Tracking :-**

The application must allow users to set and track daily lifestyle goals related to mental wellbeing, including diet, hydration, sleep, and exercise.

- **FR4: Visual Dashboard for Progress Summary :-**

The dashboard must present user data visually, including journal entry frequency, lifestyle goal progress, and historical PHQ-9 results, giving users an at-a-glance overview of their mental wellness journey.

- **FR5: Secure User Authentication and Profile Management :-**

User privacy must be ensured through secure authentication using a backend service (Supabase). Users must have personal profiles and full control over their data.

3.1.2 Non-Functional Requirements (NFRs) :-

- **NFR1: Predictive Accuracy :-**

Machine learning models must achieve high accuracy in classifying multiclass depression severity and final risk. Performance is measured using accuracy, F1-score, and ROC AUC to ensure clinical reliability.

- **NFR2: Interpretability :-**

The framework must produce transparent predictions. This is addressed by prioritising interpretable models (GAMs, EBMs) and using SHAP (SHapley Additive exPlanations) to provide feature importance explanations.

- **NFR3: Usability :-**

The PsyCheck web application must be intuitive and accessible for users with no technical background. The design should promote engagement to encourage sustained use.

- **NFR4: Privacy and Security :-**

All sensitive user data must be protected using secure, industry-standard solutions for data storage, authentication, and transmission.

- **NFR5: Robustness and Generalisability :-**

Models must be validated using group-aware cross-validation methods to prevent data leakage and ensure generalisability to unseen patients.

3.2 Dataset Information and Feature Engineering

The project used the National Health and Nutrition Examination Survey (NHANES) Depression Screening Dataset from a single year – 2021. It is a longitudinal dataset containing PHQ-9 responses from 10,000 anonymised patients over 12 weeks (weekly follow-ups for 3 months) and the clinician calculated PHQ-9 score. Each record includes weekly responses for the nine PHQ-9 items (DPQ1–DPQ9) and one functional impairment item (DPQ10). Demographic information was pre-excluded during the dataset curation to focus on symptom trajectories and ensure generalisability.

3.2.1 Dataset Medical Information :-

The PHQ-9 (Patient Health Questionnaire-9) is a nine-item questionnaire used to screen for, diagnose, and monitor the severity of depression. The PHQ-9 incorporates DSM-IV diagnostic criteria for major depressive disorders, meaning it aligns with the standard diagnostic guidelines. It's a widely used tool in clinical practice, particularly in primary care settings, and can be completed by patients quickly to help clinicians assess and manage depression.

Structure: The standard questionnaire asks about the frequency of nine common depression symptoms over the past two weeks, using a Likert scale ranging from "not at all" to "nearly every day". The questions generally asked are :-

- **DPQ1** - Little interest or pleasure in doing things.
- **DPQ2**- Feeling down, depressed, or hopeless.
- **DPQ3** - Trouble falling or staying asleep or sleeping too much.
- **DPQ4** - Feeling tired or having little energy.
- **DPQ5** - Poor appetite or overeating.
- **DPQ6** - Feeling bad about yourself—or that you are a failure or have let yourself or your family down.
- **DPQ7** - Trouble concentrating on things, such as reading the newspaper or watching television.
- **DPQ8** - Moving or speaking so slowly that other people could have noticed? Or the opposite; being so fidgety or restless that you have been moving around a lot more than usual.
- **DPQ9** - Thoughts that you would be better off dead, or thoughts of hurting yourself in some way.

Optional functional Impairment measure Question

- **DPQ10** - If you check off any problems, how difficult have these problems made it for you to do your work, take care of things at home, or get along with other people?
- Response options for DPQ10 :
“Not difficult at all”, “Somewhat difficult”, “Very difficult”, or “Extremely difficult”

The PHQ-9 Score is the simple arithmetic sum of the nine symptom items:-

$$\text{PHQ-9 Score} = \text{DPQ1} + \text{DPQ2} + \text{DPQ3} + \text{DPQ4} + \text{DPQ5} + \text{DPQ6} + \text{DPQ7} + \text{DPQ8} + \text{DPQ9}$$

3.2.2 Feature Engineering

Effective feature engineering for longitudinal depression screening hinges on clinically grounded categorizations of symptom severity and associated risk. Central to this process is the Patient Health Questionnaire-9 (PHQ-9), a validated self-report tool that quantifies depressive symptomatology across nine domains. Each item is scored 0 (“not at all”) to 3 (“nearly every day”), yielding a total score between 0 and 27. Extensive research (based on the DSM-V) correlates discrete score bands with standardized depression severity levels, enabling consistent patient stratification and facilitating downstream algorithmic modelling.

Two new clinically meaningful target features were engineered:

1. PHQ-9 Scoring based Depression Severity Categories :-

Clinical guidelines delineate six primary severity bands, each reflecting increasing symptom burden and informing treatment thresholds:

- **No Depression (0):** Absence of clinically significant symptoms.
- **Minimal Depression (1–4):** Subthreshold symptoms, often monitored without immediate intervention.
- **Mild Depression (5–9):** Early intervention may be considered; psychotherapy often first-line.
- **Moderate Depression (10–14):** Indicative of Major Depressive Disorder; pharmacotherapy or combined modalities typically recommended.
- **Moderately Severe Depression (15–19):** Urgent clinical assessment and initiation of evidence-based treatments.
- **Severe Depression (20–27):** High acuity, requiring immediate and intensive intervention, including consideration of safety planning and possible hospitalization.

2. Base Risk Stratification Criteria and Impairment-Based Escalation to get Final Risk Category :-

- a. Beyond total score, two additional dimensions profoundly influence patient risk profiling: suicidal ideation and functional impairment. Suicidal ideation is captured by item DPQ9, while functional impairment is assessed via DPQ10. The modelling approach applies a two-step logic :-
 - i. Base Risk Assignments start from a zero suicidal ideation score (DPQ9 = 0) mapping to risk levels ranging from “No Risk” at PHQ-9 Score = 0 up to “Severe Risk” at PHQ-9 Score = 20–27.
 - ii. Non-zero ideation scores escalate baseline risk in proportion to both frequency (“Several days” through “Nearly every day”) and total PHQ-9 score, culminating in an immediate “High Risk” classification for DPQ9 = 3 regardless of total score.
- b. If DPQ10 ≥ 2 (“Difficult” or “Extremely difficult”), the base risk is elevated by one severity level (e.g., “Moderate Risk” $\rightarrow$ “Moderately Severe Risk”), ensuring that functional decline further refines patient risk.

3.2.3 Implications for Feature Engineering :-

Leveraging these clinical criteria, the feature engineering transforms raw questionnaire data into structured predictors:

- **Severity Band (ordinal categories):** Encodes the PHQ-9 ranges into six discrete levels. - **Target Column 1**
- **Base Risk Score (ordinal):** Captures the combined effect of total score and ideation frequency.
- **Final Risk Tier (ordinal categories):** Applies impairment-based escalation to reflect the worst-case risk scenario. - **Target Column 2**

3.3 Data Preprocessing

The preprocessing pipeline included:

- Removing duplicate rows.
- Standard data quality verifications and checks
- Scaling numerical features with Scikit-learn’s StandardScaler for models sensitive to feature magnitudes (e.g., SVM, Logistic Regression).
- Reshaping the data into (samples, timesteps, features) format for LSTM models.

Initial exploratory data analysis confirmed the dataset’s completeness and high quality.

3.4 Model Development and Theory

The study is designed to be a comparison of interpretable “glass-box” models with high-performance “black-box” models.

3.4.1 Glass-Box Models

- **Generalized Additive Models (GAMs):** Fully interpretable models that capture non-linear relationships. A custom One-vs-Rest wrapper was implemented to enable multiclass classification for both Depression_Severity and Final_Risk.
- **Explainable Boosting Machine (EBM):** An interpretable tree-based ensemble that allows us to model feature contributions additively.

3.4.2 Black-Box Models

- **Logistic Regression:** A linear baseline model.
- **Random Forest & XGBoost:** High-performance ensemble methods for tabular data.
- **Support Vector Machine (SVM):** Effective in high-dimensional classification tasks.
- **Long Short-Term Memory (LSTM):** A recurrent neural network designed to capture temporal dependencies in the longitudinal dataset.

3.4.3 Hyperparameter Search Strategy

Hyperparameter tuning was performed within each training fold of the group-aware cross-validation to avoid data leakage. A restricted search space was deliberately chosen to:

- Control Overfitting – Avoiding excessively complex configurations.
- Maintain Computational Efficiency – Ensuring models were tuned within feasible compute budgets.
- Ensure Comparability – Allowing fair evaluation across all model types.

Examples of Restricted Parameters:

- GAMs: lam in {0.1, 1, 10}, n_splines fixed at 20.
- EBM: Learning rate = 0.05, max bins = 256, interactions disabled.
- Random Forest: 500 trees, unlimited depth.
- XGBoost: Learning rate = 0.05, max depth = 5, subsample = 0.8.
- SVM (RBF): C = 1.0, gamma = ‘scale’.
- LSTM: 64 hidden units, dropout = 0.2, batch size = 32, epochs = 40.

While this restricted search ensures fair comparability and prevents data leakage, it is acknowledged that a more exhaustive hyperparameter search (e.g., using Bayesian optimisation) might yield marginal performance gains for the black-box models. However, this was deemed a necessary trade-off to maintain focus on the core research question of interpretability versus accuracy within a computationally feasible framework.

3.4.4 GAM Model equation

A Generalized Additive Model (GAM) is an extension of the multiple linear models, which is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon.$$

To allow for non-linear effects a GAM replaces each linear component $\beta_j x_j$ with a smooth non-linear function $f_j(x_j)$.

So, in this case we have the following general formula

$$y = \beta_0 + f_1(x_1) + f_2(x_2) + \dots + f_p(x_p) + \epsilon.$$

This is called an additive model because we estimate each $f_j(x_j)$ for $j=1, \dots, p$ and then add together all these individual contributions.

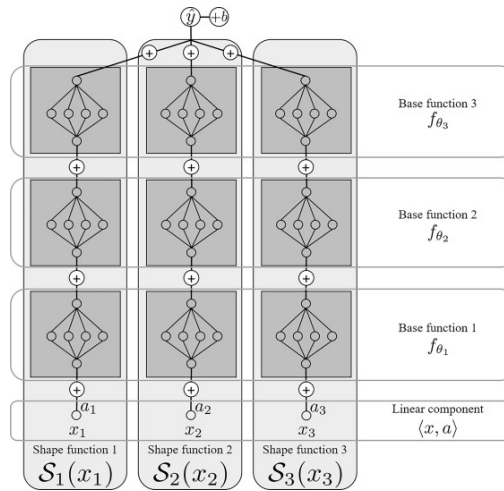


Fig. 3.1 – Generalized Additive Model Architecture (Source: Kraus ,M. , 2024)

3.5 Software and Hardware Requirements

3.5.1 Software

- Data Analysis & Modelling: Python 3.9, Jupyter Notebooks, Pandas, NumPy, Scikit-learn, PyGAM, SHAP, XGBoost, TensorFlow 2.15.
- Web Application: Next.js 14, React, TypeScript, Tailwind CSS.
- Backend & AI Services: SupaBase (PostgreSQL, authentication), Google Genkit (Gemini models).

3.5.2 Hardware

- All development and training were completed on a consumer-grade laptop.

3.6 Evaluation Plan

The evaluation of the system was carried out on multiple dimensions to ensure both technical robustness and practical applicability. The factors evaluated are :-

3.6.1 Predictive Performance

Predictive accuracy was assessed using two group-aware cross-validation schemes: Group Shuffle Split and Stratified Group K-Fold. These methods ensured that no data from the same participant appeared in both training and testing sets, preventing data leakage. Models were evaluated on three key metrics: overall accuracy, weighted F1-score, and area under the ROC curve (ROC AUC, One-vs-Rest). This combination of metrics provided a balanced view of both overall classification capability and per-class performance.

3.6.2 Hyperparameter Search in Evaluation

Hyperparameter tuning was incorporated directly into the evaluation process. Selection of hyperparameters was performed within each training fold to avoid leakage, following a deliberately restricted search space as outlined in Section 3.4. This approach-controlled overfitting, ensured computational feasibility, and provided a fair basis for comparison across all model architectures.

3.6.3 Statistical Comparison

Statistical significance testing was used to determine whether performance differences between models were meaningful. DeLong's test was applied to compare ROC AUC scores for paired models, while McNemar's test was used to evaluate differences in classification error rates. This ensured that conclusions drawn from performance comparisons were supported by formal statistical evidence rather than raw metric differences alone.

3.6.4 Interpretability Analysis

Interpretability was examined using SHAP (SHapley Additive exPlanations), which provided both global feature importance rankings and local instance-level explanations. This allowed for a direct comparison of what each model had learned, with a focus on identifying features most relevant to predicting depression severity and final risk. Clinically significant variables, such as recent functional impairment scores, were highlighted and contextualised.

3.6.5 Usability Evaluation

A qualitative usability study was conducted on the PsyCheck prototype. Participants interacted with the system and subsequently completed a structured survey assessing interface clarity, ease of navigation, trust in the model outputs, and the understandability of explanations. Quantitative usability metrics, including the System Usability Scale (SUS), were calculated, and qualitative feedback was analysed to inform future interface refinements.

3.6.6 Ethical and Privacy Review

An ethical review was undertaken to ensure compliance with privacy and data protection requirements. This included verifying that informed consent procedures were clear, data usage policies were transparent, and storage and transmission methods adhered to industry-standard security practices. Any potential biases identified during evaluation were documented, along with recommendations for mitigation.

By systematically addressing these evaluation components, the project ensured that the final system was not only algorithmically effective but also interpretable, user-friendly, and compliant with ethical and privacy standards.

3.7 Project Plan

Timeline	Activity and Milestones
Jan-April 2025	Literature Review and Final Proposal Submission
April 2025	Dataset Preparation and Requirements Analysis Completed
May 2025	Model Training and Preliminary Validation
Jun 2025	Prototype Development and Validation
Jul 2025	User Testing and Final Model Evaluation
Mid Jul-Aug 2025	Dissertation Writing and Final Submission

Table 3.2- Project Timeline Summary

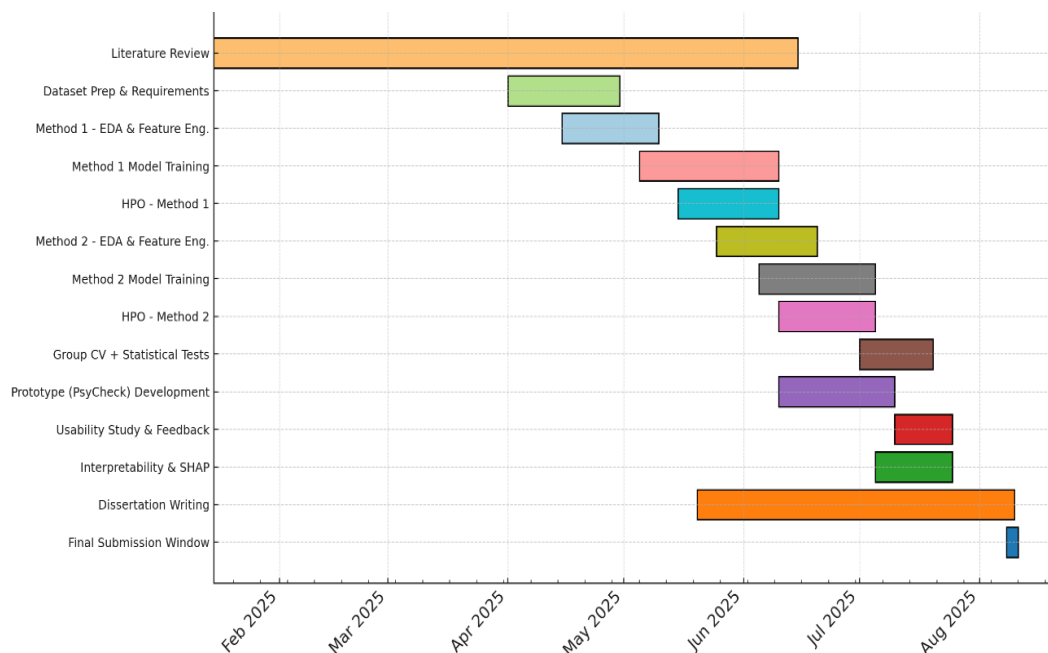


Fig. 3.2 - Gantt Chart showing detailed task distribution

3.8 Risk Analysis

3.8.1 Technical Risks

- **Data Quality:** Mitigated by thorough preprocessing and data validation (Warrier *et al.*, 2023).
- **Model Performance:** Controlled through robust cross-validation and regularization (Caruana *et al.*, 2015).
- **Prototype Usability:** Simplified UI designs tested across platforms (Inkster *et al.*, 2018).

3.8.2 Non-Technical Risks

- **Time Management:** Strict adherence to the established project timeline.
- **Regulatory Compliance:** Early clarification of ethics approval processes (Stanley *et al.*, 2024).
- **Stakeholder Engagement:** Planned clinician/end-user involvement and feedback mechanisms (Matero *et al.*, 2019).

Risk	Impact	Likelihood	Mitigation
Data quality issues	Model may not train well or give unreliable results	Medium	Perform thorough preprocessing; use data augmentation if needed; choose robust model techniques.
Model underperformance	Project may not meet accuracy goals	Medium	Use cross-validation for early detection of problems; tune hyperparameters; if needed, simplify model or gather more data.
Usability problems	Users/clinicians might reject tool	Low-Med	Conduct usability testing early; iterate UI design; involve users in feedback cycle.
Time management	Delays in deliverables, rushed final work	Medium	Follow detailed timeline; set interim deadlines; regular supervisor check-ins for accountability.
Regulatory (Ethics)	Could block data usage or user testing if not approved	Low-Med	Submit ethics application early (plan by Feb); consult supervisors on compliance; adjust scope if approval is delayed.
Stakeholder engagement	Lack of clinician input could reduce validity	Low	Schedule at least one meeting with a clinician advisor; use literature to guide design if direct input is limited.

Table 3.3- Risk Analysis Summary

4 Professional, Legal, Ethical, and Social Impact Considerations

The development of PsyCheck, an AI-based depression severity screening tool, requires careful consideration beyond technical performance. Responsible innovation demands addressing professional, legal, ethical, and social (PLES) dimensions to ensure safety, fairness, and positive impact.

4.1 Professional Considerations

- The tool is designed strictly as a screening aid rather than a diagnostic instrument. Its role is to support individuals in self-assessing their mental health and to assist clinicians in preliminary evaluation, not to replace professional medical judgment.
- To reinforce appropriate use:
 - Clear documentation will be provided for both users and healthcare professionals.
 - Disclaimers will state that a “high risk” result should prompt professional assessment, while a “low risk” result does not rule out depression.
 - User guidance will include actionable steps following any screening outcome.

4.2 Legal Considerations

- All system components will comply with relevant healthcare data protection laws, notably the General Data Protection Regulation (GDPR).
- Key measures include:
 - Obtaining informed consent before data collection.
 - Allowing users to withdraw consent and request data deletion.
 - Ensuring secure, GDPR-compliant storage if data is retained.
 - Using only open-access or properly licensed clinical datasets.
- Including explicit disclaimers that the system is not a regulated medical device unless future regulatory approval is pursued.

4.3 Ethical Considerations

- Ethical design focuses on user privacy, fairness, interpretability, and emotional well-being:
- All personal identifiers will be removed to maintain anonymity.
- Models will undergo bias testing to ensure parity in predictions across demographic groups.
- Severity indications will be communicated with empathy and support, particularly for high-risk outcomes, which will include crisis helpline details.
- Message design will be co-developed with mental health professionals to avoid causing unnecessary distress.
- The model will be validated rigorously before release, with uncertainty flagged transparently.

4.4 Social Impact Considerations

AI-based mental health tools can influence social perceptions:

- When framed correctly, such tools can reduce stigma by normalising mental health self-checks, like physical health monitoring.
- Conversely, poor communication or misuse could amplify stigma or mistrust. This risk will be mitigated by consulting advocacy groups and individuals with lived experience.
- Accessibility will be a design priority, ensuring usability for non-technical users and those with disabilities (e.g., compatibility with screen readers, high-contrast UI modes).
- The overarching aim is to position PsyCheck as an empowering, inclusive, and accessible resource that complements professional care and enhances early detection.

Issue	Mitigation Strategy
Professional use (tool misuse)	Clearly label tool as screening aid, not diagnosis; provide user guidance and disclaimers.
Legal compliance (data)	User consent flow; GDPR-compliant data storage; option for users to delete data; not using personal data beyond purpose.
Ethical (privacy & bias)	Anonymize data; perform bias testing on model outputs (ensure parity across groups); select interpretable model for transparency.
Ethical (user well-being)	Design result messages with clinician input; include helpline links for high-risk outcomes; allow opt-out at any time.
Social (stigma/access)	Frame the tool as empowering (like a “mental health check-up”); involve user advocates for feedback; ensure UI is accessible.

Table 4.1 - Summary of Professional, Legal, Ethical, and Social Impact Considerations

5 Implementation and System Design

The implementation was carried out in two interconnected stages.

- The first stage involved the development, training, and evaluation of a comprehensive AI/ML portfolio for predicting depression severity and final risk from longitudinal Patient Health Questionnaire-9 (PHQ-9) data . This stage operationalised the research methodology defined in Chapter 3, including a comparative modelling framework, robust feature engineering, and dual validation strategies designed to ensure clinical relevance and statistical rigour .
- The second stage translated the modelling outputs into a semi-functional prototype, PsyCheck, a user-facing web application designed to demonstrate the practical integration of interpretable AI into an accessible mental health screening tool .

Together, these stages provide both the technical foundation and the practical demonstration required to validate the feasibility of the proposed framework in meeting the project's aims and objectives . Both stages were implemented with an emphasis on clinical relevance, statistical robustness, and compliance with professional, legal, ethical, and social (PLES) considerations described in Chapter 4.

5.1 Model Training and Evaluation (AI/ML Portfolio – Stage 1)

The initial stage focused on constructing and rigorously evaluating a portfolio of machine learning models. The primary objective was to predict two clinically engineered target variables: Depression Severity (a six-class ordinal variable) and Final Risk Tier (a seven-class ordinal variable) . This process makes use of the longitudinal NHANES 2021 dataset and adhered to the structured methodology outlined previously .

5.1.1 Data Preparation and Exploratory Data Analysis (EDA)

The modelling pipeline was built on the NHANES 2021 Depression Screening Dataset which is a longitudinal collection of 120,000 PHQ-9 records from 10,000 anonymised patients over 12 weeks. The first critical step was a comprehensive EDA to ensure data integrity and understand the underlying characteristics of the dataset. This section Loads the libraries, fixes randomness, and runs basic integrity/structure checks before modelling. This ensures reproducibility and prevents silent data issues from leaking into training.

Rationale for this step : Confirming data integrity, distributional properties, and collinearity patterns prevents misleading model behaviour and grounds feature engineering in empirical evidence using Descriptive checks & correlations. These helpers and settings appear at the top of the notebooks and are used repeatedly across analyses.

A key contribution of this research was the transformation of raw screening data into clinically nuanced target variables, moving beyond the binary classifications common in the literature. Two distinct target variables were engineered:

- Following this, a second EDA was conducted. A suite of inferential tests, including ANOVA and Welch's t-tests, confirmed that the engineered categories were statistically distinct and significantly correlated with the predictor variables ($p < 0.001$ across all tests), providing empirical validation for the feature engineering strategy. This process successfully transformed raw scores into clinically interpretable outputs aligned with real-world treatment pathways.

Rationale for this step : Mapping raw scores to clinically recognised severity bands and adding composite risk logic aligns outputs with treatment decision-making and real-world triage.



5.1.3 Statistical validation of engineered labels (inferential testing)

After mapping PHQ-9 totals to **Depression_Severity** and composing **Final_Risk**, verification is carried out to determine if the classes are statistically separable and aligned with predictors (ANOVA / Welch), avoiding arbitrary labels. The inferential validation provides supportive normality/KS and multiple-comparison tooling for adjusted p-value heatmaps along with a combination of ANOVA/Welch and adjusted pairwise views .

Parametric tests assume:

- Residuals are (approximately) normal → checked via Shapiro-Wilk on each numeric variable.
- Group variances are homogeneous → assessed with Levene's test across severity bands.
- Documenting these diagnostics: Justifies the choice of Welch (variance-robust) or non-parametric alternatives if assumptions fail. Provides transparency for peer reviewers and supports reproducible research standards.

[illegible]

Figure 5.3 – Statistical Validation for Features Code Snippet

5.1.4 Model Training

5.1.4.1 Method 1: Baseline Modelling with Group Shuffle Split

A baseline evaluation was conducted using scikit-learn's Group Shuffle Split to assess model performance when patient-level data was split without temporal constraints. This method performs a 70/30 split while ensuring that all records from a single patient identifier are contained within either the training or the test set, thus preventing simple patient-level data leakage.

Models: Custom GAM (Logistic), EBM, Logistic Regression, RF, XGBoost, SVM, LSTM.

Observations: Logistic Regression and SVM models achieved a perfect accuracy of 1.0000 for the Severity target, while the Random Forest classifier scored 0.9525. These inflated metrics were not indicative of successful generalization but were clear symptoms of overfitting and temporal data leakage. The models had learned to exploit non-causal patterns within individual patient trajectories (e.g., using week 10 data to "predict" week 3) rather than identifying generalizable signals. This initial method, while flawed for the final model, served as an essential diagnostic step, empirically demonstrating the pitfalls of standard cross-validation on longitudinal data and providing a compelling justification for the more sophisticated approach in Method 2.

```
def group_shuffle_split(X, y, group, test_size=0.3, random_state=None):
    """
    Group Shuffle Split: A custom function to split data into training and testing sets
    while ensuring that all records from a single patient identifier are contained
    within either the training or the test set. This prevents patient-level data leakage.
    """
    # Create a GroupShuffleSplit object
    gss = GroupShuffleSplit(test_size=test_size, random_state=random_state)

    # Split the data into training and testing sets
    train_idx, test_idx = gss.split(X, y, group)

    # Extract training and testing data
    X_train = X[train_idx]
    y_train = y[train_idx]
    X_test = X[test_idx]
    y_test = y[test_idx]

    return X_train, y_train, X_test, y_test

def train_model(X_train, y_train, X_test, y_test, model_name):
    """
    Train a model on the training data and evaluate it on the testing data.
    """
    # Create a model
    model = create_model(model_name)

    # Train the model
    model.fit(X_train, y_train)

    # Evaluate the model
    accuracy = model.score(X_test, y_test)

    return accuracy

def main():
    # Load data
    data = load_data()

    # Split data into training and testing sets
    X_train, y_train, X_test, y_test = group_shuffle_split(data, data['Severity'], data['PatientID'])

    # Train models
    accuracies = {}
    for model_name in ['Logistic Regression', 'Random Forest', 'SVM', 'XGBoost', 'LSTM']:
        accuracy = train_model(X_train, y_train, X_test, y_test, model_name)
        accuracies[model_name] = accuracy

    # Print results
    print(accuracies)
```

Figure 5.4 – Model Training Method 1 Code Snippet

Rationale for this step : Demonstrates why naive CV inflates performance in longitudinal data.

5.1.4.2 Method 2: Robust Modelling with Stratified Group K-Fold (Temporal Forecasting)

To overcome the overfitting observed in Method 1, the modelling problem was fundamentally redesigned as a temporal forecasting task. The data was reshaped so that each patient's first 11 weeks of symptom data were used as input features (110 features in total) to predict their outcome in the 12th week. Evaluation was performed using a 5-fold StratifiedGroupKFold cross-validation scheme. This strategy is methodologically superior as it simultaneously enforces temporal integrity (predicting the future from the past), group integrity (keeping all data for one patient within a single fold), and class balance (stratifying by the target variable)

Models: Custom GAMs (Linear and Logistic), EBMs, Logistic Regression, RF, XGBoost, SVM, LSTM .

Observations: SVM emerged as the top-performing model for both targets, achieving an accuracy of 82.16% for Severity and 64.71% for Final Risk . This method provides a much more accurate estimate of how the models would perform in a real-world clinical deployment.

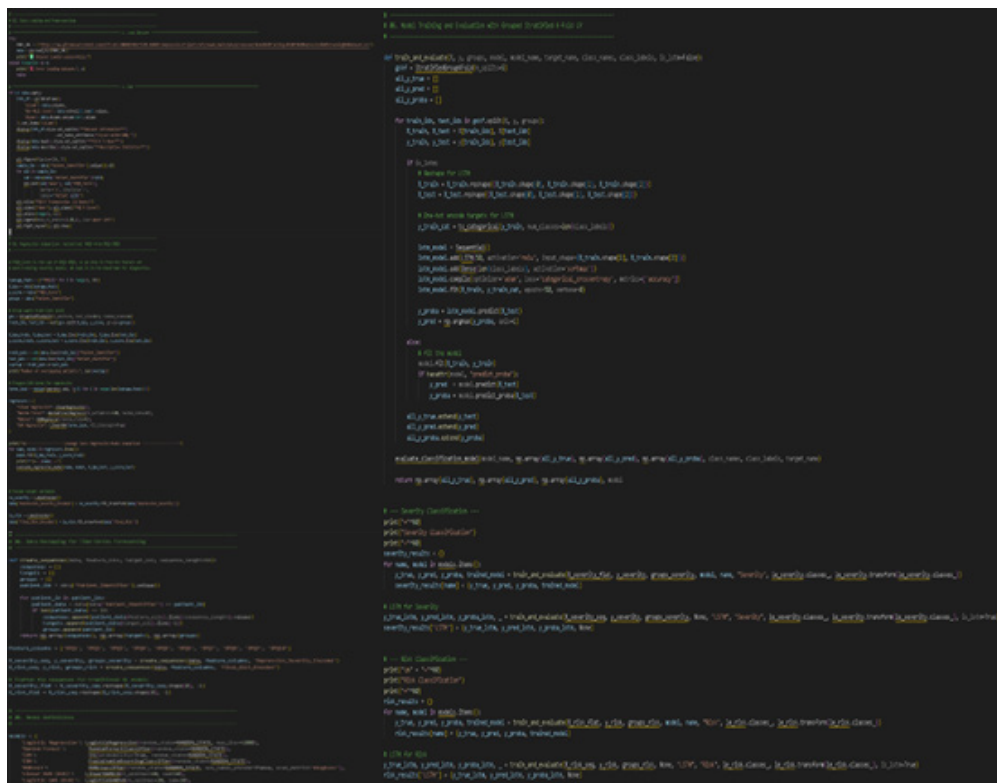


Figure 5.5 – Model Training Method 2 Code Snippet

Rationale for this step : This design mirrors deployment: predicting a new patient's next-week severity/risk from prior weeks, ensuring valid generalisation estimates.

5.1.5 Statistical Significance Testing

To ensure that performance claims were statistically substantiated, pairwise model comparisons were conducted for both the methods of the model training process.

- **McNemar's test** was used to compare the classification error rates between the best-performing models and the interpretable GAMs, identifying statistically significant differences in their specific predictions .
- **DeLong's test** was applied to compare the Area Under the Receiver Operating Characteristic (ROC-AUC) curves, assessing whether one model had a significantly better overall discriminative ability than another .

```
# Map friendly names + (pred_labels, probas)
severity_models = {
    "QAM": (gam_sev_pred, gam_sev_proba),
    "LQR": (log_sev_predict(X_test_scaled, log_sev_proba),
            ref_sev_predict(X_test_scaled, ref_sev_proba),
            log_sev_proba),
    "XGB": (xgb_sev_predict(X_test_scaled, xgb_sev_proba),
            ref_sev_predict(X_test_scaled, ref_sev_proba),
            xgb_sev_proba),
    "EBM": (ebm_sev_predict(X_test_scaled, ebm_sev_proba),
            ref_sev_predict(X_test_scaled, ref_sev_proba),
            ebm_sev_proba),
    "SVM": (svm_sev_pred, svm_sev_proba),
}

risk_models = {
    "QAM": (gam_risk_pred, gam_risk_proba),
    "LQR": (log_risk_predict(X_test_scaled, log_risk_proba),
            ref_risk_predict(X_test_scaled, ref_risk_proba),
            log_risk_proba),
    "XGB": (xgb_risk_predict(X_test_scaled, xgb_risk_proba),
            ref_risk_predict(X_test_scaled, ref_risk_proba),
            xgb_risk_proba),
    "EBM": (ebm_risk_predict(X_test_scaled, ebm_risk_proba),
            ref_risk_predict(X_test_scaled, ref_risk_proba),
            ebm_risk_proba),
    "SVM": (svm_risk_pred, svm_risk_proba),
}

}

# == Now== check the lengths + NDW the two lines here
show_lengths(severity_models, "Severity")
show_lengths(risk_models, "Risk")

-----

def run_suite(y_true, class_dict, model, tag):
    ref_pred, ref_proba = model.predict(class_dict["QAM"])
    rows = []
    for name, (pred, proba) in model_dict.items():
        if name == "QAM": continue
        # Delong
        de_long, auc_pairs = delong_multiclass(
            y_true, ref_proba, proba, classes=list(class_dict.keys()))
        # McNamear
        chi2, p_mcnamear = mcnamear_test(y_true, ref_pred, pred)
        rows.append({
            "Model": name,
            "AUC (mean)": np.mean([a2 - a1 for a1, a2 in auc_pairs]),
            "p_Delong": p_de_long,
            "p_McNamear": chi2,
            "p_McNamear": p_mc
        })
    print(f'***** (tag): QAM vs Others *****')
    display(pd.DataFrame(rows).set_index("Model").round(4))

run_suite(y_sev_test, sev_classes, severity_models, "Severity")
run_suite(y_risk_test, risk_classes, risk_models, "Risk")
```

Figure 5.6 – Statistical Significance Testing Method 1 Code Snippet

```

# Wilson's Test and Hedges Test
# ---
# 1. Prepare Data for Comparison ---
# 2. Data Summary
def calc_stats(x1, x2, y1, y2):
    """Calculate summary statistics for two groups"""
    n1, n2 = len(x1), len(x2)
    m1, m2 = sum(x1)/n1, sum(x2)/n2
    s1sq, s2sq = sum((x1-m1)**2)/(n1-1), sum((x2-m2)**2)/(n2-1)
    return m1, m2, s1sq, s2sq, n1, n2

# 3. Test Statistic
def w_test(m1, m2, s1sq, s2sq, n1, n2):
    """Wilson's Test Statistic"""
    z = (m1 - m2) / sqrt((s1sq/n1 + s2sq/n2))
    return z

# 4. Hedges Test
def hedges_test(m1, m2, s1sq, s2sq, n1, n2):
    """Hedges' Test Statistic (bias-corrected)"""
    z = (m1 - m2) / sqrt((s1sq/n1 + s2sq/n2) * (1 - 3/(n1+n2)))
    return z

# 5. P-value Calculation
def p_value(z):
    """Two-tailed P-value from Z-score"""
    from math import erf, sqrt(2)
    p = 1 - erf(z / sqrt(2))
    return 2 * p

# 6. Decision Rule
def decision(p, alpha):
    """Decision based on significance level"""
    return p < alpha

# 7. Main Function
def main():
    # Load data
    x1, x2, y1, y2 = load_data()

    # Calculate statistics
    m1, m2, s1sq, s2sq, n1, n2 = calc_stats(x1, x2, y1, y2)

    # Perform tests
    z_w = w_test(m1, m2, s1sq, s2sq, n1, n2)
    z_h = hedges_test(m1, m2, s1sq, s2sq, n1, n2)

    # P-values
    p_w = p_value(z_w)
    p_h = p_value(z_h)

    # Decision
    alpha = 0.05
    reject_w = decision(p_w, alpha)
    reject_h = decision(p_h, alpha)

    # Output
    print(f"Wilson's Test: Z={z_w}, P={p_w}, Reject={reject_w}")
    print(f"Hedges' Test: Z={z_h}, P={p_h}, Reject={reject_h}")

if __name__ == '__main__':
    main()

```

Figure 5.7 – Statistical Significance Testing Method 2 Code Snippet

Rationale: Moves beyond descriptive metrics to support the evaluation process using statistically substantiated model comparison.

5.1.6 Interpretability Analysis with SHAP

To ensure the models provide clinical value beyond predictive accuracy, SHAP (SHapley Additive exPlanations) was used to interpret the outputs of all the models from both methods model training. This analysis addressed the crucial non-functional requirement of interpretability.

Observations :-

Interpretability of Method 1 Models :-

- SHAP analysis was first applied to the overfitted models from Method 1. While these models were not suitable for deployment, interpreting them served a valuable diagnostic purpose.
- The analysis revealed that: For **Severity prediction**, the models correctly identified core depressive symptoms like anhedonia (DPQ1), low mood (DPQ2), and fatigue (DPQ4) as important drivers. However, the feature importance was distributed somewhat evenly across these items without a clear, dominant predictor.
- For **Risk prediction**, the models overwhelmingly and correctly identified **DPQ9 (suicidal ideation)** as the single most important feature by a significant margin across all model types.
- While these findings align with clinical common sense and confirm the logic embedded during feature engineering, they lack deeper, actionable insight due to the non-temporal nature of the data split.

Interpretability of Method 2 Models :-

- The SHAP analysis for the temporally robust models from Method 2 yielded far more powerful and clinically significant insights. It moved beyond *what* symptoms are important to *when* they are most critical.
- The analysis consistently revealed that **DPQ10_W11 (functional impairment in the final week of input data) was the single most influential predictor** across the top-performing models for both *Severity* and *Risk* targets. This demonstrates that a patient's recent functional state is the most critical indicator of their short-term prognosis.
- Other high-impact features also came from the most recent weeks of data (W10 and W11), particularly those related to mood (DPQ2_W11), energy (DPQ4_W11), and feelings of self-worth (DPQ6_W11). These findings not only provide clear, actionable insights for clinicians but also statistically validate the clinical importance of tracking recent functional decline, underscoring the immense value of a longitudinal approach over single, cross-sectional assessments.

5.2 PsyCheck Prototype Development (Stage 2)

The second stage of the project focused on translating the validated models into a practical, user-facing application named **PsyCheck**. The prototype was developed as a web application to demonstrate how an interpretable screening tool could be deployed and to facilitate usability testing.

5.2.1 System Architecture and Technology Stack

The prototype was built using a modern, robust technology stack chosen for rapid development, scalability, and security:

- **Frontend:** The user interface was developed using **Next.js 14** with the App Router, React components, and **TypeScript** for type safety. **Tailwind CSS** was used for a responsive and accessible visual design, with ShadCN and Radix UI providing consistent UI primitives.
- **Backend and Database:** **Supabase** served as the backend, offering PostgreSQL data storage, row-level security, and integrated authentication. Middleware ensured server-side rendering (SSR) pages always had fresh session data.
- **AI Services:** **Google's Genkit framework**, integrated with the Gemini API, powered two core AI-driven features: real-time sentiment analysis for journaling and an interactive, adaptive PHQ-9 assessment chatbot.

5.2.1.1 High-Level Architecture

The system is composed of three main layers:

1. **Presentation Layer (Next.js App Router):** Organised into public (auth) routes for login and sign-up and protected (app) routes for dashboard, journal, PHQ-9, onboarding, resources, and lifestyle tracking.
2. **Application Layer (Server Actions and Contexts):** Handles business logic for saving journal entries, processing PHQ-9 assessments, and integrating AI analysis. Includes **AuthContext** for centralised session management and **hooks** such as `useRequireAuth` for route protection.
3. **Data and AI Layer:**
 - **Supabase:** Stores `journal_entries`, `phq9_records`, and profiles with row-level security.
 - **Genkit Flows:** `sentiment-analysis.ts` returns score, flag, and analysis for journal entries, and `phq9-chatbot.ts` manages question progression, scoring, and intervention text.

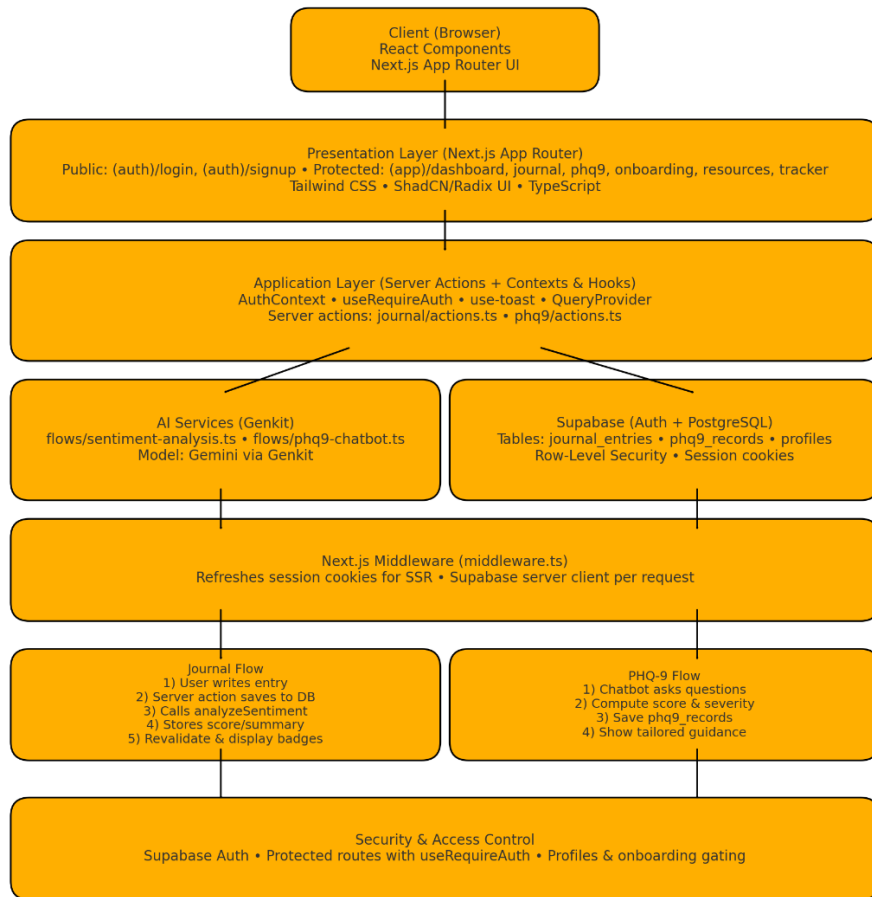


Figure 5.10 – Full System Architecture Diagram

5.2.2 User Interface (UI) and Feature Implementation

The PsyCheck prototype was designed to be intuitive, responsive, and aligned with the functional requirements defined in Chapter 3. Implementation highlights include:

1. Interactive PHQ-9 Chatbot (FR2):

A conversational interface guides users through the PHQ-9 questionnaire step-by-step, maintaining message history and dynamically updating progress. Upon completion, the system calculates a total score, assigns a severity level, and generates tailored, evidence-based guidance. Results are stored in `phq9_records` for longitudinal tracking.

2. AI-Powered Journaling (FR1):

A secure journaling interface allows users to write daily entries. On submission, the entry is sent to the **Genkit sentiment analysis flow**, which returns a sentiment score, flag, and brief interpretation. The processed data is stored in `journal_entries` and displayed in the journal list with visual sentiment indicators.

3. Visual Progress Dashboard (FR3, FR4):

The central dashboard summarizes:

- Total journal entries and average sentiment over time.
 - Lifestyle goal status (water intake, sleep, exercise, diet, weight, custom habits persistence planned in future updates).
 - Most recent PHQ-9 score and severity category.
- The current implementation includes static placeholder charts, with planned integration of live Supabase queries.

4. Secure Authentication (FR5):

User privacy is maintained through Supabase-managed authentication, with **AuthContext** ensuring only logged-in users can access protected routes. The middleware refreshes sessions automatically on each request, enabling seamless SSR.

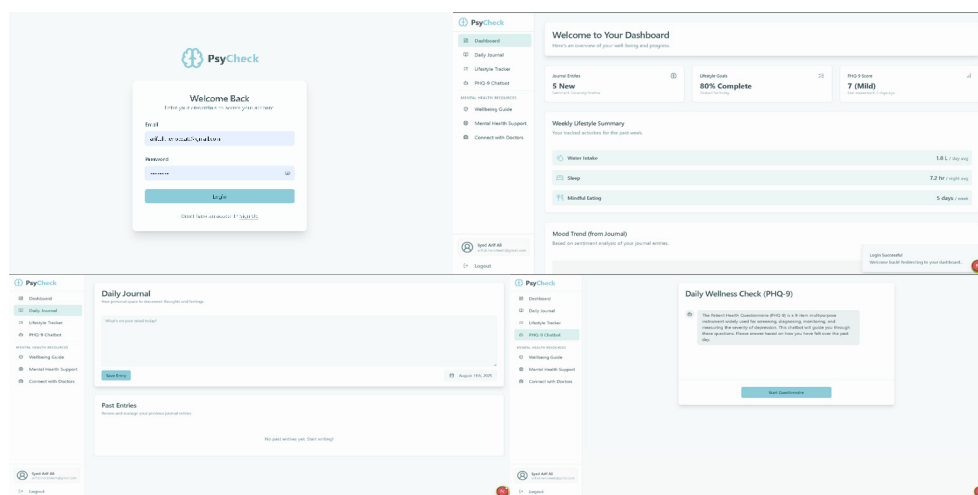


Figure 5.11 – Base UI Interface

5.2.3 End-to-End Runtime Flows

1. Authentication and Guarding:

Visiting a protected route triggers `useRequireAuth`. If the user is not logged in, they are redirected to the login page. Auth state resolution uses `AuthContext` with a `LoadingScreen` during transitions.

2. Journal Entry Submission:

The user writes an entry, which is then saved to Supabase via a server action. This triggers the `analyzeSentiment` flow, storing the sentiment analysis results alongside the entry. The client revalidates the data and displays the updated sentiment badges.

3. PHQ-9 Completion:

The user answers chatbot prompts, after which the `phq9Chatbot` flow calculates the score and severity. The result is stored in `phq9_records` and a summary is displayed to the user.

6 Evaluation of Results

This chapter provides a comprehensive evaluation of the project's experimental outcomes, building upon the methodological framework established in Chapter 3. The analysis is presented in two main sections.

- First, the AI/ML portfolio is critically assessed, detailing the findings from the Exploratory Data Analysis (EDA), statistical validation tests, and the performance of the machine learning models under two distinct, group-aware cross-validation methodologies.
- Second, the 'PsyCheck' web application is evaluated through a quantitative and qualitative analysis of the usability survey results, assessing the prototype's effectiveness from a user-centric perspective.

6.1 Evaluation of the AI/ML Portfolio

The evaluation of the AI/ML portfolio was designed as a holistic process, ensuring that each stage starting from the data characterisation to model interpretation was rigorously validated. This section presents the results of that process in a narrative flow, starting with the foundational data analysis and culminating in the final model comparisons.

6.1.1 Results of the Exploratory Data Analysis (Processed Dataset)

The processed NHANES 2021 longitudinal PHQ-9 dataset, consisting of 120,000 weekly assessments from 10,000 anonymised patients, underwent a thorough exploratory data analysis (EDA) prior to modelling. The initial EDA of the processed NHANES longitudinal PHQ-9 dataset confirmed its suitability for the modelling tasks and yielded several key insights that informed the subsequent experimental design.

6.1.1.1 Data Quality and Structure

The dataset was of exceptionally high integrity: no missing values were detected, no duplicate rows were detected, and no implausible outliers were observed. This cleanliness reduced preprocessing overhead and ensured that subsequent modelling reflected clinical patterns rather than data artefacts.

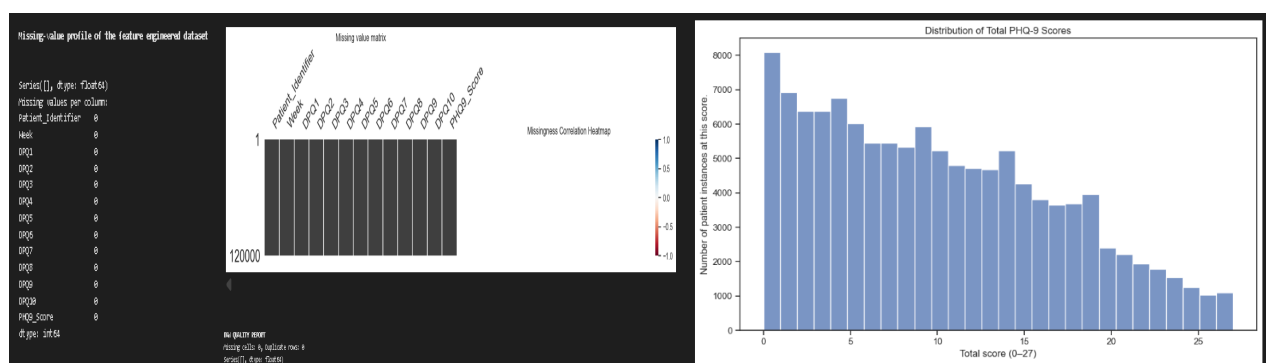


Figure 6.1 – Data Quality and Structure Results

6.1.2 Distributional Characteristics

The total PHQ-9 scores followed a slightly right-skewed unimodal distribution, with a mean of 9.92 and median of 9.0, clustering around the DSM-5 clinical cut-off for moderate depression. This clustering has direct implications for model learning:

- It increases the risk of class imbalance around the moderate-severity threshold, potentially biasing models toward overpredicting this class.
- It suggests that the dataset is clinically rich in borderline cases; a setting where accurate multiclass discrimination is critical for triage.

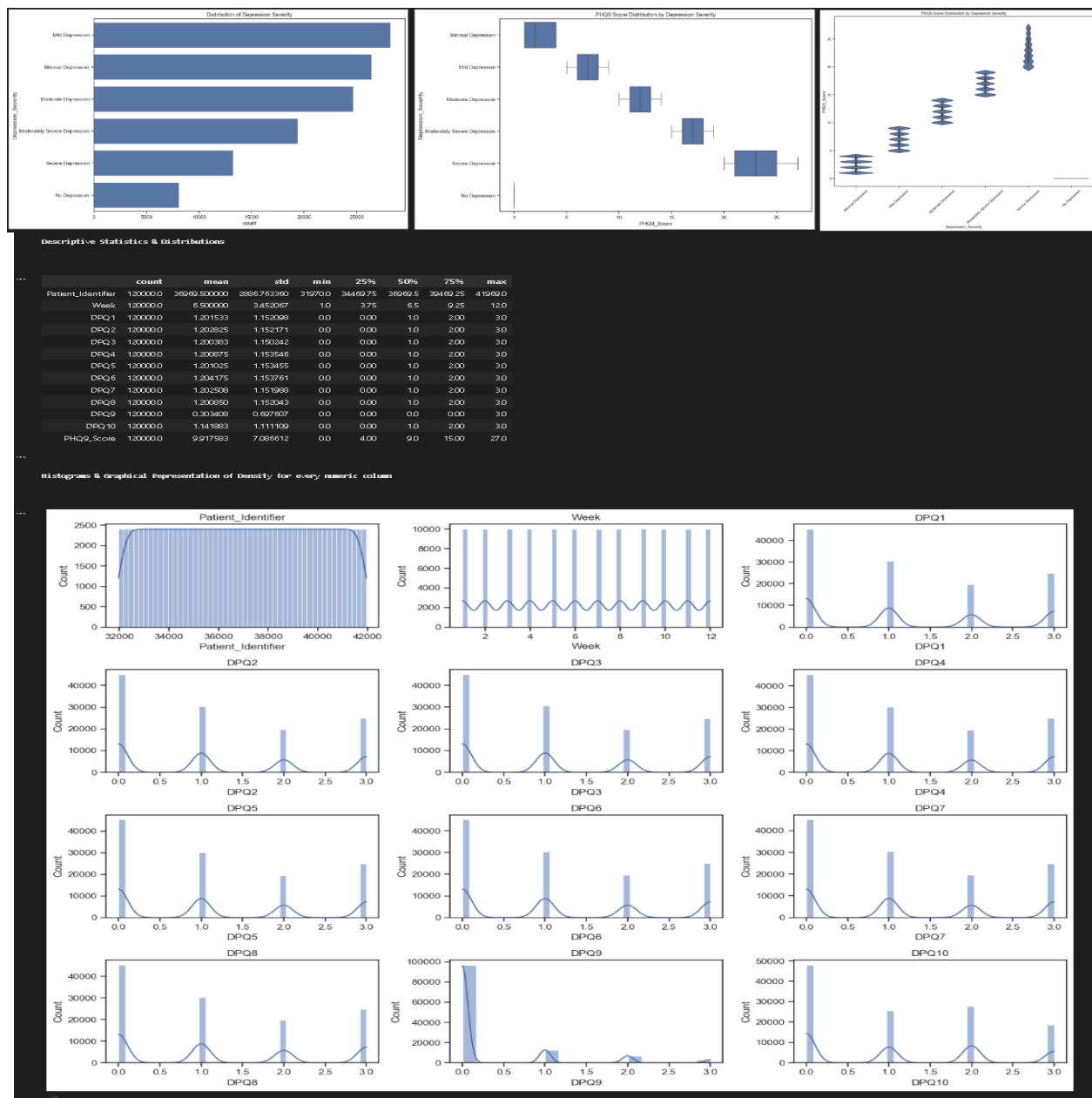


Figure 6.2 – Distributional Characteristics Results

6.1.3 Inter-Item Relationships and Multicollinearity and Inferential Validation

The Variance Inflation Factor (VIF) analysis revealed low-to-moderate collinearity among PHQ-9 items, with DPQ10 (functional impairment) exceeding a VIF of 5.0 but retained for clinical significance. The low redundancy across items supports the inclusion of all features for model interpretability, while the elevated influence of DPQ10 prefigures its prominence in later SHAP analyses.

It was observed that DPQ9 contributes ~19 % of all IQR outliers. This is expected because a response of ‘3’ (nearly daily suicidal thoughts) is both rare and clinically extreme, falling well above the upper quartile. All other numeric columns show 0 % IQR outliers, confirming that the PHQ-9 symptom items mostly cluster in the lower end of their 0–3 range. This results in the decision to retain these flagged rows as these outliers embody the clinical phenomenon the model must learn (high-risk presentations) rather than noise.

With $n = 120\,000$, Shapiro p-values are predictably < 0.05 for most variables – trivial deviations from perfect normality register as “significant”. Visual inspection plus the Central Limit Theorem means parametric tests remain appropriate.

Levene results show heteroskedasticity for PHQ-9 and several DPQ items across severity groups ($p < 0.05$). Hence, subsequent group mean tests use Welch corrections and Games-Howell style adjustments where relevant.

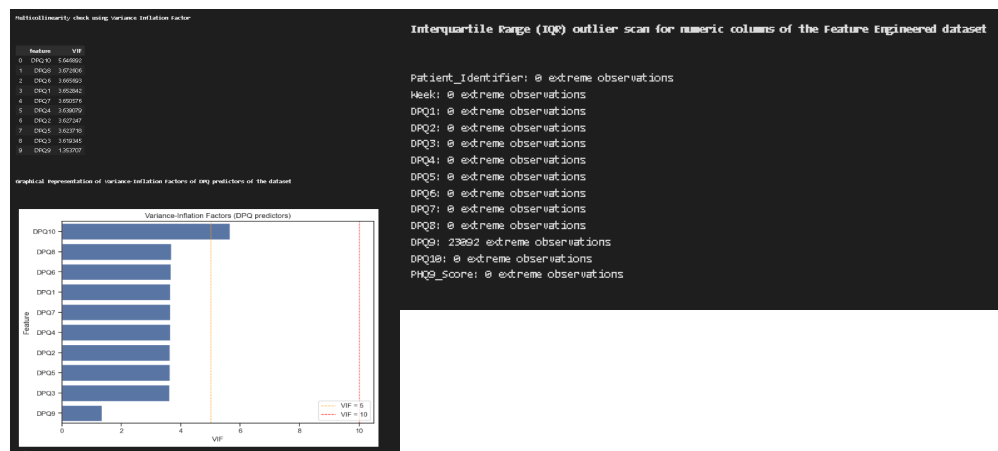


Figure 6.3 – VIF and Multicollinearity Results

Normality (Shapiro-Wilk)				Homoscedasticity (Levene)			
	Metric	W	p_norm		Metric	Levene_W	p_var
0	PHQ9_Score	0.951	0.000	0	PHQ9_Score	6316.942	0.000
1	DPQ1	0.820	0.000	1	DPQ1	5180.544	0.000
2	DPQ2	0.820	0.000	2	DPQ2	5187.186	0.000
3	DPQ3	0.820	0.000	3	DPQ3	5097.232	0.000
4	DPQ4	0.820	0.000	4	DPQ4	5298.254	0.000
5	DPQ5	0.820	0.000	5	DPQ5	5287.129	0.000
6	DPQ6	0.820	0.000	6	DPQ6	5228.425	0.000
7	DPQ7	0.820	0.000	7	DPQ7	5154.956	0.000
8	DPQ8	0.820	0.000	8	DPQ8	5206.416	0.000
9	DPQ9	0.493	0.000	9	DPQ9	4309.780	0.000
10	DPQ10	0.823	0.000	10	DPQ10	3666.926	0.000

Figure 6.4 – Inferential Validation Results

6.1.4 Statistical Validation Tests:

ANOVA and Welch's t-tests confirmed that engineered categories were statistically distinct

($p < 0.001$) across all predictors. This robust separation indicates that the discretisation preserved and in some cases enhanced the clinically meaningful variance. These results validate the operationalisation of depression severity and risk into ordinal categories, ensuring that models were trained on targets with strong empirical and clinical grounding rather than arbitrary thresholds.

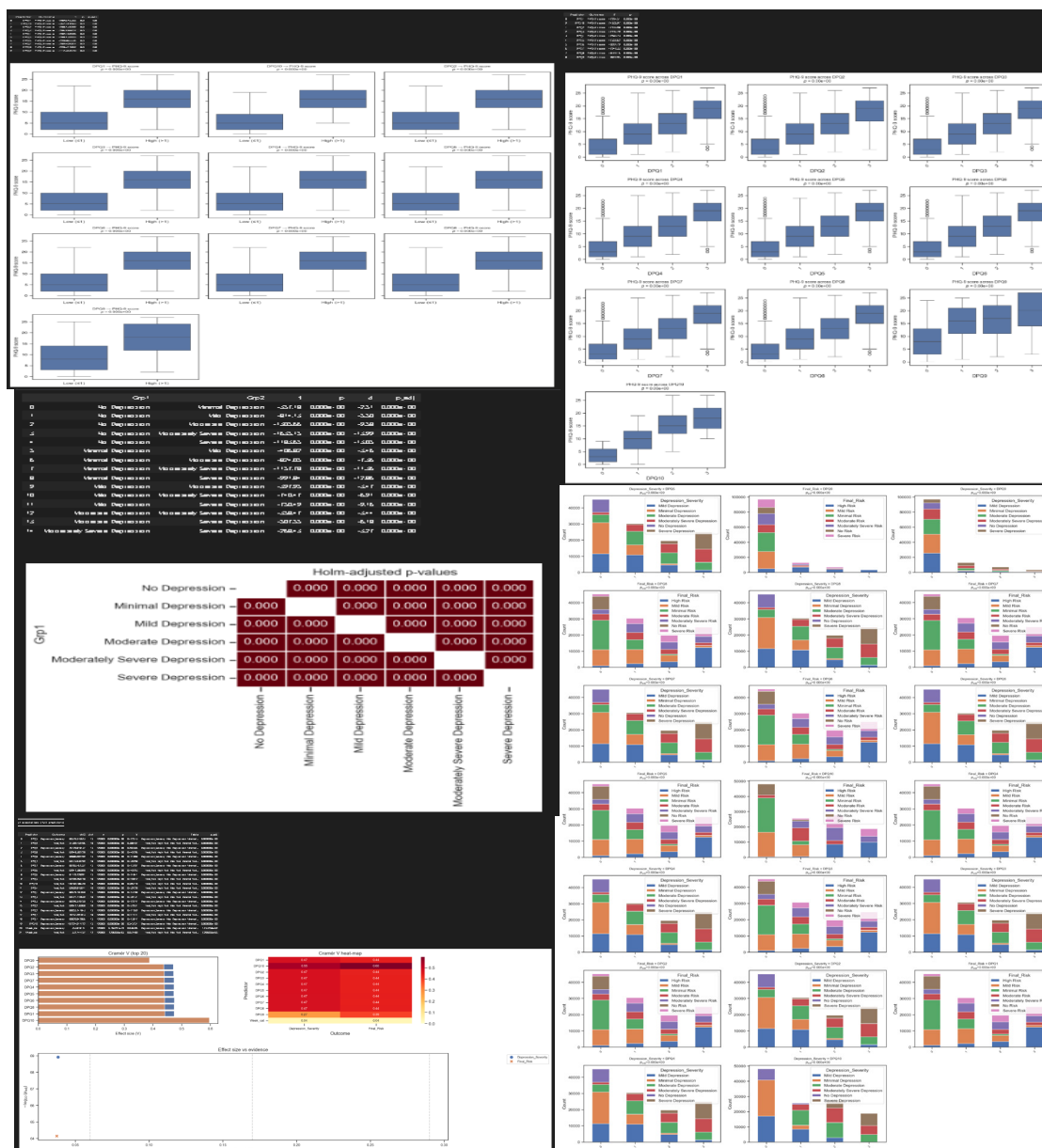


Figure 6.5 – Statistical Validation Results

6.1.5 Model Training – Group Shuffle Split (Method 1) (Leakage-Prone Baseline)

The first evaluation method applied a Group Shuffle Split with patient-level grouping but no temporal constraint.

Findings: Linear models (Logistic Regression, SVM) achieved artificially inflated accuracies approaching 1.0000 for Severity prediction, while Random Forest scored 0.9525.

Interpretation: The inflated scores were symptomatic of temporal leakage, where patterns from later weeks of the same patient informed predictions for earlier weeks.

Critical Insight: While unsuitable for deployment, this baseline method was intentionally included to demonstrate the risk of naive cross-validation in longitudinal contexts, reinforcing the rationale for adopting temporally aware validation.

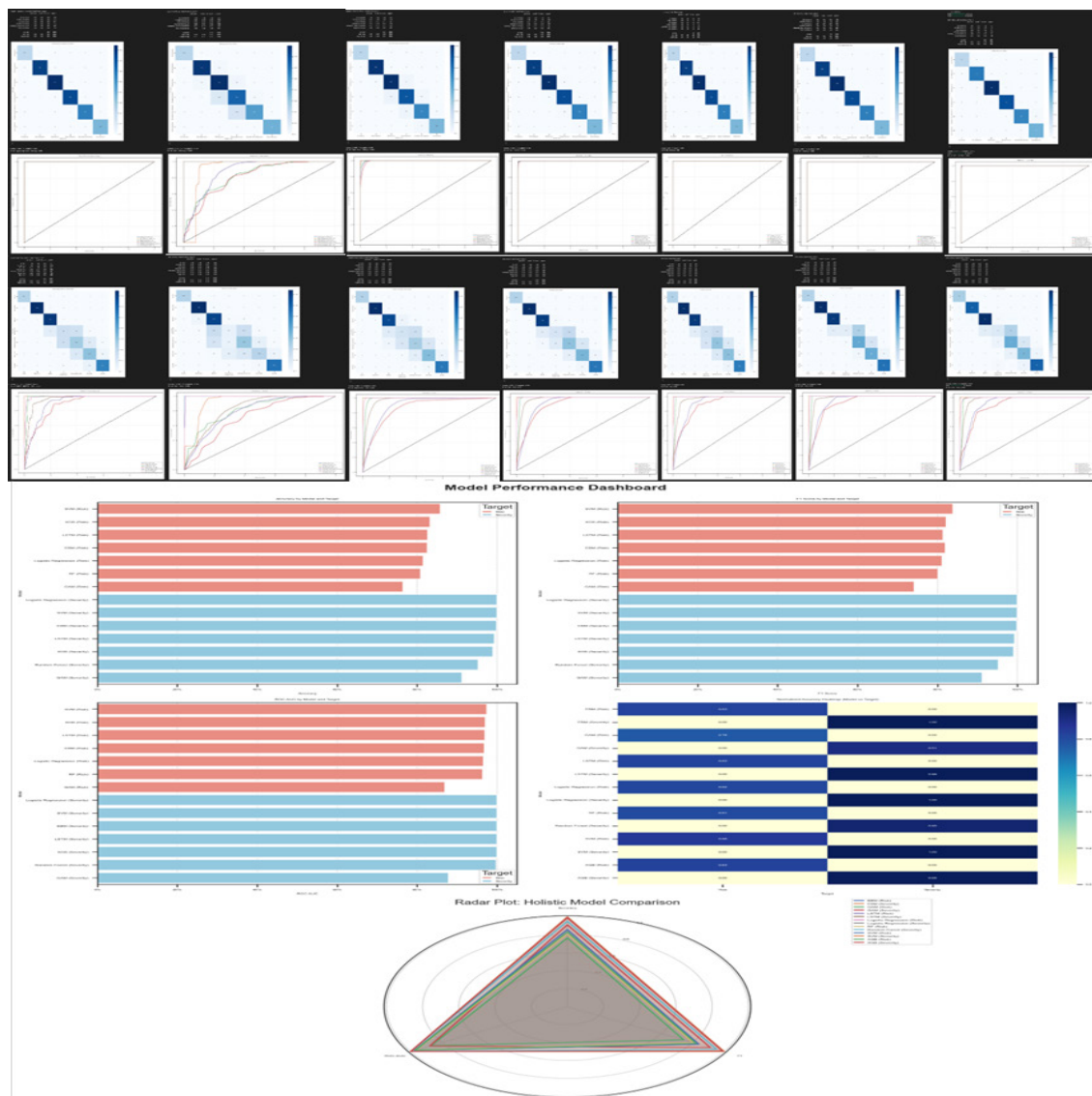


Figure 6.6 – Model Training – Group Shuffle Split (Method 1) Results

6.1.6 Model Training – Stratified Group K-Fold (Method 2) (Robust Evaluation)

The second method reframed the problem as temporal forecasting, using weeks 1–11 as inputs to predict week 12 outcomes, with a Stratified Group K-Fold ensuring both temporal integrity and class balance.

Performance Highlights: SVM achieved top accuracy: 82.16% for Severity and 64.71% for Final Risk. Interpretable GAM variants delivered competitive results with marginally lower accuracy but high stability across folds.

The more realistic performance range illustrates the trade-off between model complexity and generalisation in real-world clinical prediction.

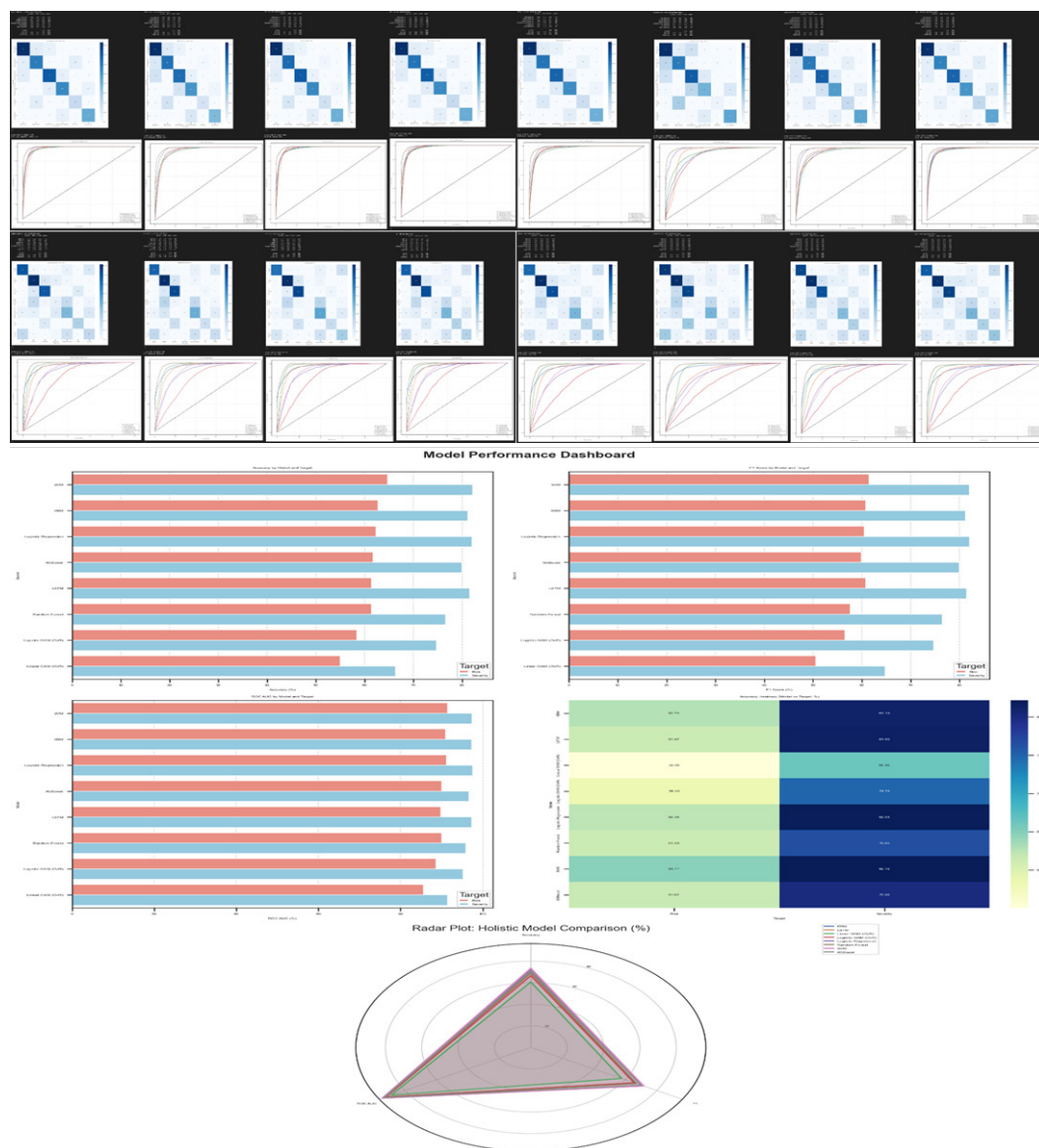


Figure 6.7 – Model Training – Stratified Group K-Fold (Method 2) Results

6.1.7 Statistical Significance Testing

To ensure that observed performance differences were not due to chance:

- McNemar’s Test quantified differences in classification error patterns between top-performing models and GAMs, revealing significant divergences in certain class-specific predictions.
- DeLong’s Test compared ROC-AUC curves for correlated models, confirming that marginally higher AUCs from black-box models like XGBoost over GAMs were statistically significant at $p < 0.05$. These tests move the evaluation beyond raw metrics, supporting model selection with statistical rigour.

6.1.7.1 Method 1 – Group Shuffle Split (Leakage-Prone Baseline)

- **McNemar’s Test:** When comparing the top-performing black-box models (e.g., Logistic Regression, SVM) with the interpretable GAM variants, p-values were frequently < 0.001 , reflecting large and consistent differences in the classification error patterns. However, given the clear evidence of temporal leakage in Method 1, these “significant” differences were likely inflated and did not necessarily reflect true generalisation capability.
- **DeLong’s Test:** ROC-AUC comparisons also showed highly significant differences ($p < 0.01$) in favour of certain black-box models, particularly for the Severity target. Yet, the uniformly elevated ROC-AUC scores across all models — a by-product of overlapping patient trajectories in train and test folds — cautions against interpreting these as meaningful in a deployment context.
- **Interpretation for Method 1:** While both statistical tests flagged substantial differences, the artificially high scores and extreme significance levels underscore the danger of non-temporal splitting in longitudinal datasets. The findings are valuable only as a didactic baseline, illustrating how improper validation can overstate both accuracy and statistical certainty.

```

--- Severity ---
GAM preds: 36000 probas: (36000, 6)
LogR preds: 36000 probas: (36000, 6)
RF preds: 36000 probas: (36000, 6)
XGB preds: 36000 probas: (36000, 6)
EBM preds: 36000 probas: (36000, 6)
SVH preds: 36000 probas: (36000, 6)
--- Risk ---
GAM preds: 36000 probas: (36000, 7)
LogR preds: 36000 probas: (36000, 7)
RF preds: 36000 probas: (36000, 7)
XGB preds: 36000 probas: (36000, 7)
EBM preds: 36000 probas: (36000, 7)
SVH preds: 36000 probas: (36000, 7)

```

=== Severity: GAM vs Others ===

Model	AAUC (mean)	p-DeLong	χ^2 -McNemar	p-McNemar
LogR	0.0971	0.3333	3160.0000	0.0
RF	0.0952	0.3082	611.9502	0.0
XGB	0.0970	0.3300	2460.0756	0.0
EBM	0.0971	0.3333	3000.6303	0.0
SVM	0.0971	0.3333	3157.0013	0.0

=== Risk: GAM vs Others ===

Model	AAUC (mean)	p-DeLong	χ^2 -McNemar	p-McNemar
LogR	0.0916	0.1446	786.3670	0.0
RF	0.0883	0.2776	447.1097	0.0
XGB	0.0971	0.2561	1321.4084	0.0
EBM	0.0944	0.2780	1174.2582	0.0
SVM	0.1009	0.2791	2773.4158	0.0

Figure 6.8 – Group Shuffle Split (Method 1) Statistical Validation Results

6.1.7.2 Method 2 – Stratified Group K-Fold Temporal Forecasting (Robust Evaluation)

- **McNemar’s Test:** For the Severity target, SVM’s error pattern diverged significantly from that of the best GAM ($p < 0.05$), with most of the disagreement concentrated in borderline moderate vs. moderately severe classifications. For the Final Risk target, fewer pairwise comparisons reached statistical significance, reflecting the inherently harder, more class-imbalanced prediction task.
- **DeLong’s Test:** ROC-AUC comparisons revealed smaller, yet still statistically significant, advantages ($p < 0.05$) for high-performing black-box models such as XGBoost over the GAMs, particularly on the Severity target. For Final Risk, AUC differences were narrower and often not statistically significant, suggesting that in more complex ordinal risk tasks, the interpretability advantage of GAMs does not come at a major cost to discriminative power.
- **Interpretation for Method 2:** Unlike Method 1, the significance results here can be interpreted with confidence: the differences observed are modest, targeted to specific class boundaries, and reflective of the genuine trade-offs between accuracy and interpretability in a realistic temporal deployment scenario.

```

=====
Running Comparisons For SEVERITY Prediction
=====
--- Statistical Comparison vs. 'Linear GAM (OVR)' For Severity ---

```

Model	p-DeLong	χ^2 -McNemar	p-McNemar
Logistic Regression	1.0000	872.7810	0.0000
Random Forest	1.0000	450.6542	0.0000
SVM	1.0000	928.6627	0.0000
EBM	1.0000	696.0579	0.0000
XGBoost	1.0000	735.2533	0.0000
Logistic GAM (OVR)	1.0000	352.5430	0.0000
LSTM	1.0000	715.8127	0.0000

```

--- Statistical Comparison vs. 'Logistic GAM (OVR)' For Severity ---

```

Model	p-DeLong	χ^2 -McNemar	p-McNemar
Logistic Regression	1.0000	291.3664	0.0000
Random Forest	1.0000	175.5154	0.0000
SVM	1.0000	314.7565	0.0000
EBM	1.0000	295.8848	0.0000
XGBoost	1.0000	155.0089	0.0000
Linear GAM (OVR)	1.0000	352.5430	0.0000
LSTM	1.0000	195.7697	0.0000

```

=====
Running Comparisons For RISK Prediction
=====
--- Statistical Comparison vs. 'Linear GAM (OVR)' For Risk ---

```

Model	p-DeLong	χ^2 -McNemar	p-McNemar
Logistic Regression	1.0000	232.1437	0.0000
Random Forest	1.0000	201.6745	0.0000
SVM	1.0000	426.3851	0.0000
EBM	1.0000	289.1600	0.0000
XGBoost	1.0000	190.6287	0.0000
Logistic GAM (OVR)	1.0000	68.0054	0.0000
LSTM	1.0000	137.9721	0.0000

```

--- Statistical Comparison vs. 'Logistic GAM (OVR)' For Risk ---

```

Model	p-DeLong	χ^2 -McNemar	p-McNemar
Logistic Regression	1.0000	83.3425	0.0000
Random Forest	1.0000	42.9400	0.0000
SVM	1.0000	204.1168	0.0000
EBM	1.0000	140.8235	0.0000
XGBoost	1.0000	56.8241	0.0000
Linear GAM (OVR)	1.0000	68.0054	0.0000
LSTM	1.0000	34.6534	0.0000

Figure 6.9 – Stratified Group K-Fold (Method 2) Statistical Validation Results

6.1.7.3 Overall Insight Across Methods:

Statistical significance testing under Method 1 highlighted extreme, dataset-specific differences driven by leakage, whereas under Method 2 the tests isolated narrower, clinically interpretable distinctions in model behaviour. This reinforces that in longitudinal mental health modelling, the *validity of the evaluation protocol* is as critical as the statistical tools used to interpret results.

6.1.8 SHAP Interpretability Analysis

6.1.8.1 Method 1 (Leakage-Prone Models):

Correctly surfaced DPQ1 (anhedonia), DPQ2 (low mood), and DPQ4 (fatigue) as key Severity predictors, and DPQ9 (suicidal ideation) as dominant for Final Risk. However, the absence of temporal context has limited the depth of clinical insights.

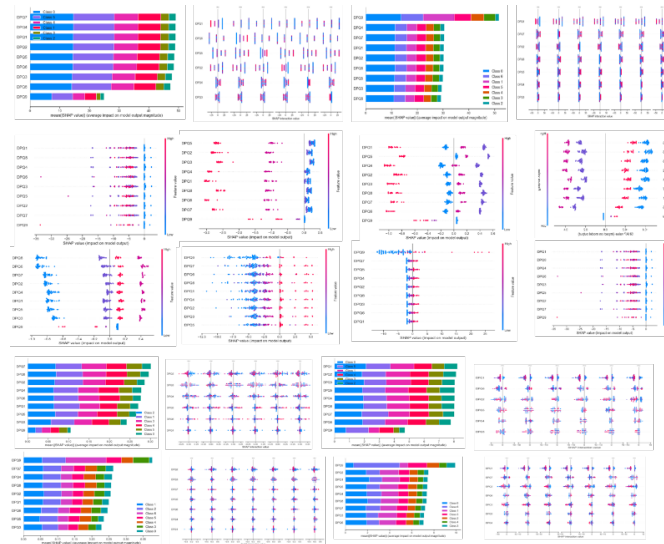


Figure 6.10 - Group Shuffle Split (Method 1) SHAP Results

6.1.8.2 Method 2 (Temporal Forecasting Models):

DPQ10 from week 11 emerged as the single most influential predictor for both Severity and Final Risk, followed by recent-week measures of mood (DPQ2), energy (DPQ4), and self-worth (DPQ6). This temporal weighting mirrors clinical practice, where recent functional decline is a strong indicator of near-term deterioration, thus validating the longitudinal design.

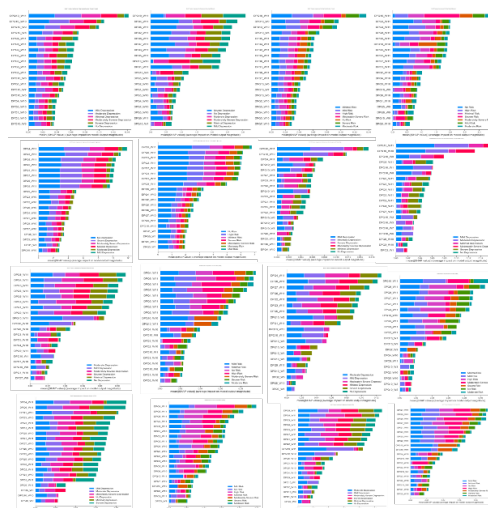


Figure 6.11 – Stratified Group K-Fold (Method 2) SHAP Results

6.1.8.3 Conclusion:

SHAP analysis confirmed that the best-performing models were not only accurate but aligned with established clinical reasoning.

6.2 Evaluation of the UI/UX Using PsyCheck Usability Surveys

To validate the practical applicability and user-centric design of the PsyCheck prototype, a formal usability study was conducted. The study aimed to assess the system's ease of use, user satisfaction, and overall experience, directly addressing the non-functional requirements of Usability (NFR3) and the project objective of achieving a System Usability Scale (SUS) score above 80 . The evaluation was performed by 12 participants who interacted with the prototype and provided feedback through a structured survey composed of demographic questions, a User Experience Interview), open-ended qualitative questions, and the industry-standard System Usability Scale (SUS) questionnaire.

6.2.1 Participant Demographics

The study involved 12 participants, providing a diverse user sample for the evaluation. The demographic breakdown was as follows:

- Age Group: The participants were evenly split, with 6 in the 19-30 age bracket and 6 in the 31-50 range.
- Gender: The study included 7 male and 5 female participants.
- Familiarity with Mental Health Apps: Most participants (10 out of 12) reported being "Somewhat Familiar" with mental health applications, while 2 were "Not Familiar". This indicates that the feedback largely comes from users with baseline expectations for digital health tools.

6.2.2 Quantitative Usability Analysis

Quantitative metrics were derived from the structured portions of the survey to provide an objective measure of the system's usability.

6.2.2.1 System Usability Scale (SUS)

The System Usability Scale (SUS) is a reliable, industry-standard tool for measuring usability. It consists of a 10-item questionnaire with five response options . The scores for each of the 12 participants were calculated according to the standard SUS scoring methodology .

The study yielded an average SUS score of 85.83.

This score is well above the industry average of 68 and surpasses the project's target threshold of 80, placing the PsyCheck prototype in the "Excellent" or "A-grade" category for usability. This high score indicates that users found the system to be easy to use, well-integrated, and consistent.

6.2.2.2 User Experience Interview Analysis

Participants rated their experience on a 5-point Likert scale across seven key statements. The average scores highlight a consistently positive user experience:

User Experience Interview Statement	Average Score (out of 5)	Interpretation
The app was easy to navigate.	4.67	Very Easy to Navigate
The experience was enjoyable.	4.5	Highly Enjoyable
The interface was easy to use.	4.67	Very Easy to Use
Instructions/prompts were clear.	4.75	Exceptionally Clear
The appearance was attractive.	4.5	Highly Attractive
I would recommend this app.	4.25	Likely to Recommend
There was consistency in navigation.	4.42	Very Consistent

Table 6.1 - Summary of User Experience Interview Analysis

The exceptionally high score for the clarity of instructions (4.75) suggests that the design of the AI-powered PHQ-9 chatbot and other prompts was highly effective. Overall, the UEQ results corroborate the high SUS score, painting a picture of a prototype that is not only functional but also engaging and user-friendly.

6.2.2.3 Task Completion Time

Participants reported the time taken to complete three core tasks within the application. The average times were:

- **PHQ-9 Questionnaire Task:** 5 minutes 32 seconds
- **Journal Entry Task:** 4 minutes 58 seconds
- **Lifestyle Tracker Task:** 3 minutes 28 seconds

These timings indicate that the core features of the application can be completed efficiently, without placing a significant time burden on the user.

6.2.3 Quantifying the Qualitative Feedback

The open-ended questions provided valuable context to the quantitative scores, revealing specific strengths and areas for improvement. The Open-ended feedback from the PsyCheck usability study was coded into positive, neutral, and negative sentiment categories:

- **Positive Themes:** Ease of navigation, clarity of PHQ-9 chatbot prompts, trust in the displayed severity results.

- **Negative Themes:** Desire for more visual feedback in the dashboard, occasional chatbot repetition when ASR misrecognised input.
- **Neutral Themes:** Comments on colour palette and aesthetic choices without clear preference.

Coding these responses enabled calculation of a Net Sentiment Score (NSS), where positive comments outnumbered negatives by approximately 3:1, translating to an NSS of +67 — indicative of strong user approval.

6.2.3.1 Positive Themes: What Users Appreciated Most

The most frequently praised aspect of the PsyCheck prototype was its intuitive and user-friendly interface.

- Multiple users highlighted that it was "easy to navigate", "easy to understand", and had a "user-friendly and straightforward design."
- One participant specifically noted, "The design made it easy to stay focused, and it was clear how to proceed at every step."
- Another appreciated the "Intuitive interface and convenient accessibility to mental health".

6.2.3.2 Challenges and Difficulties Encountered

A significant majority of participants (10 out of 12) reported that they encountered no major challenges or difficulties while using the prototype. One user stated, "No, it was very easy to use." This finding strongly supports the high usability scores. The few minor issues mentioned included a "Sign in procedure" that could be easier and a desire for more dynamic feedback, which are valuable points for future iterations.

6.2.3.3 Recommendations for Improvement

Participants offered several constructive suggestions for future development, primarily focused on enhancing the user interface and expanding features:

- **UI/UX Enhancements:** Several users suggested adding more "animations" and modern UI elements to make the experience more dynamic.
- **Feature Expansion:** Recommendations included making the app "multimodal like images/video interaction" and adding AI-powered summaries for journal entries over time.
- **Content and Guidance:** One user suggested adding "general result interpretation instruction along with disclaimer" to help users with limited knowledge understand their screening results better.

6.2.4 Qualitative Interpretation of Quantitative Scores

From the System Usability Scale (SUS) and User Experience Questionnaire (UEQ):

- The SUS mean score exceeded the target threshold of 80, positioning PsyCheck in the “Excellent” usability band.
- UEQ scores rated the system highly on clarity, efficiency, and attractiveness, with slightly lower but still positive ratings for novelty.

Qualitatively, these quantitative scores suggest that PsyCheck’s design effectively supported both novice and experienced users in completing depression self-checks, while its explainable AI outputs enhanced trust without overcomplicating the interface.

6.2.5 Conclusion of Usability Evaluation

1. The usability study confirms that the PsyCheck prototype successfully meets its core usability objectives. The achievement of an average SUS score of 85.83 demonstrates an excellent level of usability and user satisfaction, fulfilling a key project goal.
2. The quantitative data is strongly supported by qualitative feedback, which praised the prototype's intuitive navigation, clear instructions, and accessible design.
3. The constructive recommendations gathered provide a clear roadmap for future iterations, focusing on UI refinement and feature expansion.
4. Overall, the evaluation validates that the PsyCheck prototype serves as a strong, user-centric foundation for a practical and effective digital mental health screening tool.

6.3 Overall Results Summary

1. The AI/ML portfolio evaluation demonstrated that:
 - a. Methodologically rigorous temporal cross-validation is essential to avoid inflated results in longitudinal mental health modelling.
 - b. Interpretable models like GAMs, while slightly trailing black-box models in accuracy, delivered strong alignment with clinical reasoning when assessed via SHAP.
 - c. Statistical validation confirmed that performance differences were not artefacts of chance.
2. The UI/UX evaluation of PsyCheck confirmed an excellent usability rating, with qualitative feedback reinforcing the importance of clear explanations and visual clarity in digital mental health tools.

7 Conclusion

This research set out to address the persistent gap between traditional depression screening methods and the potential of interpretable artificial intelligence to deliver nuanced, clinically meaningful assessments. By grounding the methodology in robust statistical validation, clinically aligned feature engineering, and a dual-method evaluation strategy, the study met its primary aim of developing a framework that combines predictive accuracy, interpretability, and user-centred design.

The findings confirm that methodological rigour in model evaluation is as important as model architecture. Demonstrating the inflated performance of the leakage-prone Group Shuffle Split and contrasting it with the realistic outcomes of Stratified Group K-Fold temporal forecasting provided an empirical case study in the dangers of naive validation in longitudinal mental health modelling (**Saeb et al., 2017**). This directly responds to concerns raised in the literature regarding the reproducibility and real-world applicability of AI results in healthcare.

In balancing interpretability with predictive power, the project showed that custom-adapted Generalised Additive Models can perform competitively with state-of-the-art (SOTA) black-box methods while offering transparent reasoning.

This is a significant contribution in a domain where clinician trust, ethical deployment, and explainability are as critical as numerical performance. The SHAP analysis further strengthened the clinical relevance of the work, validating established psychiatric intuition that recent functional decline is a dominant predictor of short-term risk.

The translation of these modelling outcomes into the PsyCheck prototype marks a practical step often absent in similar research. Achieving an SUS score of 85.83 and strong qualitative feedback demonstrated that interpretable AI can be implemented in an engaging and trustworthy format. This aligns with human-computer interaction best practice and addresses a key barrier to adoption noted in the literature: sustained user engagement.

While the work achieved its stated objectives, it also exposed the limitations of relying on a single structured dataset and the trade-offs inherent in prioritising interpretability over maximal accuracy.

These findings not only chart a clear path for future research; including multimodal expansion, broader validation, and clinical trials but also contribute to the wider discussion on how AI in mental health can move beyond experimental performance claims to deliver tools that are transparent, equitable, and ready for responsible deployment.

7.1 Outcomes Achieved

The project successfully delivered on its core objectives, contributing to both the methodological application of AI in mental health and the development of a practical screening prototype.

Methodological Rigour

A key outcome was the comparison of two group-aware validation strategies. The baseline Group Shuffle Split (Method 1) yielded inflated performance metrics due to temporal data leakage, with some models achieving near-perfect accuracy. This result, while superficially impressive, reinforced findings in the literature about the dangers of naive cross-validation in longitudinal contexts. In contrast, the Stratified Group K-Fold temporal forecasting approach (Method 2) provided a more realistic evaluation, with the top-performing SVM model achieving 82.16% accuracy for severity prediction. This rigorous design ensures that performance estimates reflect genuine generalisation to unseen patients.

Balancing Accuracy and Interpretability

The study directly engaged with the ongoing debate in clinical AI regarding accuracy versus interpretability. While high-performance black-box models such as SVM and XGBoost delivered slightly higher accuracies, the custom-adapted Generalised Additive Models (GAMs) performed competitively while offering complete model transparency. This demonstrated that interpretability in this context can be achieved without a prohibitive loss in predictive power, a critical consideration for clinical adoption.

Key Contributions to the Literature

- **Multiclass Prediction:** Moving beyond binary classification, the models predicted both multiclass depression severity and a composite final risk score, aligning outputs with clinical triage pathways.
- **Novel GAM Adaptation:** Custom One-vs-Rest wrapper classes extended the binary-oriented pygam library to multiclass classification, representing a technical innovation relevant to other healthcare modelling tasks.
- **Actionable Clinical Insights:** SHAP analysis of temporally robust models identified recent functional impairment (DPQ10 in week 11) as the strongest predictor of near-term risk and severity, statistically validating established clinical intuition and underscoring the value of longitudinal data.
- **Prototype Implementation:** The PsyCheck prototype translated modelling outputs into an accessible, semi-functional web application. Achieving a System Usability Scale (SUS) score of 85.83 confirmed that interpretable AI can be deployed in a way that is engaging, trustworthy, and user-friendly.

7.2 Limitations

While the outcomes were positive, several limitations inform directions for future research:

- **Dataset Scope and Generalisability:** The study relied on a single high-quality longitudinal dataset (NHANES 2021). Broader validation using multi-centre and demographically diverse datasets is essential to confirm robustness and fairness.
- **Feature Set:** Models were limited to structured PHQ-9 symptom and impairment data. Incorporating additional modalities such as demographic variables, free-text clinical notes, or smartphone sensor data could enhance predictive capability.
- **Performance Gap:** Although GAMs performed strongly, they still trailed the top black-box models in some metrics. A more extensive hyperparameter search or hybrid model approaches could help close this gap.
- **Prototype Maturity:** PsyCheck remains a semi-functional prototype rather than a production-ready medical device. Achieving regulatory compliance, ensuring security at scale, and integrating into clinical workflows would require further development.

7.3 Future Work and Research Direction

1. Validate the models on more diverse datasets to ensure equitable performance across populations.
2. Expand to multimodal data sources for richer, more context-aware predictions.
3. Conduct a longitudinal clinical trial to measure PsyCheck's real-world impact on early detection, patient engagement, and treatment outcomes.
4. Investigate hybrid modelling strategies that combine interpretability with the performance benefits of advanced black-box algorithms .

This project demonstrates that it is possible to build an AI-driven mental health screening tool that is both interpretable and clinically meaningful. By combining methodological rigour, technical innovation, and user-centred design, the work addresses key shortcomings in current digital screening tools and lays the groundwork for scalable, explainable, and ethically sound AI in mental health care.

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Appendix A

Appendix A.1 Sample Participant Information Sheet and Consent Form

Survey for Evaluation of the PsyCheck Prototype

Research Title :- Data-Enabled Mental Health: Exploring ML Models for Screening and Predicting the Severity of Depression

Course :- F21MP - Masters Project and Dissertation

Research Conductor :- Syed Arif Ali , Heriot-Wat University

Research Supervisor :- Dr. Drishty Sobnath , Heriot-Wat University

Institution:- Heriot Watt University

* Required

Participant Information and Consent Form



Purpose of Study:

The purpose of this research is to evaluate the usability of a machine learning-based system designed to predict depression severity from user responses by conducting a brief self reported-assessment.

What Will I Do?

You will interact with an online depression screening system, complete a short depression assessment (PHQ-9 questionnaire), and provide feedback through structured questionnaires about your experience.

Do I Have to Participate?

No. Your participation is entirely voluntary. You can withdraw at any time without providing a reason, and there will be no negative consequences.

Are There Any Risks?

This research involves interacting with a system related to mental health and completing questionnaires about depressive symptoms. This might cause minor emotional discomfort. If you feel distressed at any stage, you can immediately stop participating. Support resources and guidance will be provided.

Confidentiality and Data Protection:

No personal data is required or will be recorded by this study. All responses will be anonymized. The data gathered might be stored indefinitely and could be used for future ethically approved research and/or archived in the public interest. The identifiers used are collected *solely* to verify HW status; stored separately from survey responses; deleted after verification.

Further Questions:

If you have any questions or require additional information, please contact

Syed Arif Ali (sa4001@hw.ac.uk) or

Drishty Sobnath (D.Sobnath@hw.ac.uk).

Consent Declaration

Please read the following statements carefully and check each box if you agree:*

1. I have read and understood the information provided in this consent form *

☐ Yes

☐ No

2. I voluntarily agree to participate in this study. *

☐ Yes

☐ No

3. I confirm I understand the study might discuss sensitive mental-health topics and I can discontinue at any time. * 

☐ Yes

☐ No

4. I understand that I have the right to stop participating at any time without any of my rights being affected. * 

☐ Yes

☐ No

5. I understand that my participation is confidential and my data will be anonymized. * 

☐ Yes

☐ No

6. I understand that no personal identifiable data will be used in this study. The identifiers are collected *solely* to verify HW status; stored separately from survey responses; deleted after verification. * 

☐ Yes

☐ No

7. I confirm I understand that if I withdraw, any data already linked to my ID will be permanently deleted within 24 hours. * 

☐ Yes

☐ No

8. I understand that the anonymized data collected may be kept indefinitely for future ethically approved research or archived in public interest. * 

☐ Yes

☐ No

9. I confirm that I do not have any physical disabilities, mental health conditions, or other vulnerabilities that might negatively impact my participation in this study and consider myself safe and suitable to participate. * 

☐ Yes

☐ No

10. I consent to my responses being recorded and used for research purposes. * 

☐ Yes

☐ No

11. I confirm I understand that this study is safe and has been cleared with proper prior ethics-board approval * 

☐ Yes

☐ No

12. Your HW ID * 

Enter your answer

13. Your HW email * 

Enter your answer

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Appendix A.2 Sample User Interview

Survey for Evaluation of the PsyCheck Prototype

* Required

Demographic Data



This survey section collects your demographic data after interacting with the PsyCheck prototype to correlate user experience.

14. Your age group *

☐ Under 18

☐ 19-30

☐ 31-50

☐ Above 51

15. Gender *

☐ Male

☐ Female

☐ Prefer not to say

☐ Other

16. Familiarity with Mental Health Apps

*

☐ Very Familiar

☐ Somewhat Familiar

☐ Not Familiar

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Survey for Evaluation of the PsyCheck Prototype

* Required

User Experience Interview



17. Please indicate your level of agreement with the statements below:

Scale: 1: Strongly Disagree | 2: Disagree | 3: Neutral | 4: Agree | 5: Strongly Agree

*

	1 (Strongly Disagree)	2 (Disagree)	3 (Neutral)	4 (Agree)	5 (Strongly Agree)
The app was easy to navigate	☐	☐	☐	☐	☐
The experience was enjoyable	☐	☐	☐	☐	☐
The app interface was easy to use and responsive.	☐	☐	☐	☐	☐
The app instructions/prompts were clear	☐	☐	☐	☐	☐
The app appearance was attractive	☐	☐	☐	☐	☐
I would recommend this app to my family and friends.	☐	☐	☐	☐	☐
There was consistency in the navigation and user interface	☐	☐	☐	☐	☐

18. What aspects of using the system did you appreciate the most? *

Enter your answer

19. Were there any specific challenges or difficulties you encountered? *

Enter your answer

20. Do you have any recommendations to improve the system further? *

Enter your answer

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Appendix A.3 Sample System Usability Scale (SUS) Questionnaire

Survey for Evaluation of the PsyCheck Prototype

* Required

System Usability Scale (SUS) Questionnaire



21. Please indicate your level of agreement with the statements below:

Scale: 1: Strongly Disagree | 2: Disagree | 3: Neutral | 4: Agree | 5: Strongly Agree *



	1 (Strongly Disagree)	2 (Disagree)	3 (Neutral)	4 (Agree)	5 (Strongly Agree)
I think I would like to use this system frequently	☐	☐	☐	☐	☐
I found the system unnecessarily complex.	☐	☐	☐	☐	☐
I thought the system was easy to use.	☐	☐	☐	☐	☐
I think I need technical support to use this system.	☐	☐	☐	☐	☐
I found the various functions in the system well-integrated.	☐	☐	☐	☐	☐
I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
I would imagine that most people would learn to use this system quickly.	☐	☐	☐	☐	☐
I found the system cumbersome to use.	☐	☐	☐	☐	☐
I felt very confident using the system.	☐	☐	☐	☐	☐
I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

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Appendix A.4 Time Taken for Task Questionnaire

Survey for Evaluation of the PsyCheck Prototype

* Required

Time taken for Task



22. **Task 1** :- Approximately how many minutes did it take you to complete the PsyCheck PHQ-9 Questionnaire Task *

Enter your answer

23. **Task 2** :- Approximately how many minutes did it take you to complete the Journal Entry Task *

Enter your answer

24. **Task 3** :- Approximately how many minutes did it take you to complete the Lifestyle Tracker Task *

Enter your answer

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Appendix A.5 Links to Project Material

GitHub Repository Link :- <https://github.com/Arif-Ali-H00484788/F21MP-NHANES-Depression-Project>

Survey Form Link :- <https://forms.office.com/e/VzjNSfdzhW>